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A Model for Two-Digit Number Processing Based on a Joint Garner and System Factorial Technology Analysis

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As a vital part of our daily lives, number processing has received much attention in current cognitive research. Most adults perceive with ease the numerical magnitude of a two-digit number and also the individual magnitudes of the component digits, yet the relation between the 2 is poorly understood. Application to two-digit numbers of Garner's speeded classification paradigm (Experiment 1) and of system factorial technology (Experiment 2) jointly reveals phases of independent and of coactive processing within a well-defined time-course. A new model, the interactive race coactive model, explains various facets of two-digit number processing.

Keywords: Garner interference, SFT, holistic versus componential processing

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How do people read the number *82*? A ready answer is that they read it the same way that they do the word *table*. Perhaps so, but there is a catch. Whereas the components of *table*—letters of the alphabet—are not semantic units themselves, the individual components of *82* are.[1] The component digits, *8* and *2*, carry arithmetic-semantic meaning to the same extent that does *82*, which poses a fundamental question about the psychological reality of the complete stimulus, *82*. Is *82* perceived as a unified whole, hence processed in a holistic fashion, or is it perceived merely as a combination of semantic components each of which preserves its meaning independently of the other component? There are compelling reasons to believe the latter, namely, that processing is analytic. When teaching number concepts to elementary schoolchildren, two-digit numbers are introduced by exposing their structure in a two-dimensional place-value system. This structure governs all (visual) presentations of two-digit numbers, so that "no model of numerical comparison can be purely analogical" (Dehaene, Dupoux, & Mehler, 1990, p. 626). Nevertheless, Dehaene (1997; Dotan & Dehaene, 2016) still concluded that the processing of two-digit numbers is holistic, to wit, "our brain apprehends a two-digit numeral *as a whole* and transforms it mentally into an internal quantity or magnitude . . . [so that the brain] is concerned only with numerical quantities, not with the symbols that convey them" (Dehaene, 1997, p. 76, emphasis added). The empirical evidence is divided, with some studies supporting holistic processing and some analytic processing (see Moeller, Huber, Nuerk, & Willmes, 2011; Reynvoet, Notebaert, & Van den Bussche, 2011, for reviews). Our tactic in this study was harnessing a pair of state-of-the-art tools of multidimensional processing in the quest for a resolution.

Holistic Representation of Two-Digit Numbers

The presence of the distance effect (Moyer & Landauer, 1967) with two-digit numbers provides arguably the strongest evidence for the holistic view: The larger the numerical difference between a pair of two-digit numbers, the shorter the time needed to decide which one is numerically larger (Hinrichs, Yurko, & Hu, 1981). Notably, both the left and the right numerals contribute to the observed distance effect to the extent that people ignore the shortcuts available due to the place-value structure of two-digit numbers. Thus, it took people longer to compare 59 to 65 than to compare 51 to 65, although the left digit is the same in both cases (Dehaene, 1997; Dehaene et al., 1990). This generalizes to other examples across the two-digit domain.

For further evidence, when priming numbers, crossing a decade does not affect the result (Reynvoet & Brysbaert, 1999). In a masked priming experiment, the observers named the target number with the target and the prime either crossing decades (e.g., prime 9, target 11) or present in the same decade (e.g., prime 7 and target 9). Note that in contrast to the routine numerical distance effect (Moyer & Landauer, 1967), in priming with a *naming* response, the closer the stimuli, the faster the response to the target (the typical semantic priming result; e.g., Neely, 1977). In the Reynvoet and Brysbaert (1999) experiment, presenting 7 as the

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[1] Arguably the indefinite article "a" and the pronoun "I" can sometimes (by some linguists) be considered semantic entities.

prime and 9 as the target resulted in the same priming effect as did presenting 9 as the prime and 11 as the target (moreover, cross-decade priming was symmetric, 9 primed 11 to the same extent that 11 did 9).

In another study, Brysbaert (1995) found the viewing durations for visually presented numerals in the 1–99 range to be a logarithmic function of magnitude and concluded that "the representation is [a] number line on which each integer has its unique representation" (p. 451). The notion of a "mental number line," popularized by Dehaene (1997, 2003; Zorzi, Priftis, & Umiltà, 2002), has been associated in the literature with a holistic psychological representation for two-digit numbers (Nuerk, Kaufmann, Zoppoth, & Willmes, 2004). Although this type of representation is not required by logic, the holistic nature of the mental line is rooted in Dehaene et al.'s (1990) first construal of the line as an "analogical medium . . . without access to the digital appearance of the numbers" (p. 638). The composite digits are jettisoned in mental representation, so that each two-digit number is represented by a unique analogue referent on the "holistic mental line" (Nuerk et al., 2004, p. 1199).

Concerning the distance effect again, the bulk of studies has been done with single-digit numbers (e.g., Bar, Fischer, & Algom, 2019; Cohen, 2009; van Opstal & Verguts, 2011; Verguts & van Opstal, 2005). The multidigit domain has been probed somewhat later. In this respect, the invariance observed when traversing across domains supports a unified representation. Fitousi and Algom (2006) used this tactic with respect to another classic phenomenon of numerical cognition, the size-congruity effect (SCE; Henik & Tzelgov, 1982; see also, Fitousi & Algom, 2018; Namdar, Ganel, & Algom, 2018). People select the physically larger member of a pair of numerals faster when this member is also numerically larger. Is the SCE present with two-digit numbers? To bypass the difficulty of specifying physical size with (each digit of) two-digit numbers, Fitousi and Algom (2006) represented physical size by the length of a bar positioned beneath the number stimulus. The same SCE was found for single-digit numbers and for two-digit numbers, consistent with a unique, holistic representation for two-digit numbers (see also, Algom, Dekel, & Pansky, 1996; Fitousi, 2010; Fitousi, Shaki, & Algom, 2009; but see Ganor-Stern, Tzelgov, & Ellenbogen, 2007). This body of supporting evidence granted, we next address observations that are inconsistent with the hypothesis of holistic representation.

Analytic Representation for Two-Digit Numbers

An astute observation separates all pairs of two-digit numbers into two mutually exclusive and exhaustive classes: decade-unit compatible and decade-unit incompatible. Consider the pairs 42–57 and 47–62. The numbers in both pairs are separated by a constant distance of 15, yet comparison is faster in the first pair than in the second pair. Why? The difference likely derives from compatibility: The first pair is compatible (the decades and units in each number lead to the same decision as $4 < 5$ and $2 < 7$), whereas the second pair is incompatible ($4 < 6$, but $7 > 6$). The compatibility effect offers an alternative account for data previously interpreted as supporting an overall distance effect (Nuerk, Weger, & Willmes, 2001; see also, Ganor-Stern et al., 2007). Consider again the pairs, 59–65 and 51–65. On the holistic view, the first pair is compared less rapidly than the second pair because the overall distance of 6 is smaller than the overall distance of 14. However, the first pair is incompatible (hence comparison is slow), whereas the second is compatible (hence comparison is fast). Nuerk and his coworkers (Nuerk, Weger, & Willmes, 2002; Nuerk et al., 2004) have shown that the results of virtually all two-digit comparison tasks are affected by this *compatibility effect*. Moreover, they have shown that many designs are biased so that incompatible pairs entail larger numerical differences than compatible pairs, thereby generating a false impression of a genuine holistic distance effect. In contrast to a blanket holistic representation, compatibility effects imply lexical and componential processing.

To get a better appreciation of the literature we should add that the compatibility effect is by no means a minor subordinate property of numerical comparisons (see, e.g., Nuerk & Willmes, 2005; Huber, Nuerk, Willmes, & Moeller, 2016). In point of fact, the compatibility effect poses a challenge to the holistic distance effect as an important determinant of two-digit comparisons. The compatibility effect is robust, replicates whenever it is tested, and characterized by a clearly charted developmental trajectory (Nuerk et al., 2001). Incidentally, the compatibility effect should not be confused with the classic Stroop effect (Stroop, 1935). The Stroop effect is concerned with different attributes of a single stimulus, and the observer's task is to respond to one attribute (e.g., naming the ink color), while ignoring another attribute (e.g., the carrying word). The compatibility effect, by contrast, is concerned with two different stimuli and the task is to compare them on magnitude—not to name any one component. Too, there is no particular reason to assume that all types of conflict tasks (e.g., Stroop, Simon, Affective Simon, or Flanker) ipso facto tap to the same underlying process (Chajut, Schupak, & Algom, 2009; Fitousi, 2016). Compatibility is actually derived from the same magnitude task used with the standard distance effect.

For all its impact on the magnitude-comparison literature, the compatibility effect does not exclude the presence of a distance effect for two-digit numbers (e.g., Ganor-Stern et al., 2007). In virtually all studies, compatibility does not explain the whole variance, a fact acknowledged by major exponents of the compatibility approach (e.g., Nuerk & Willmes, 2005). The typical analytic tool has been regression, and two-digit numerical distance still remains a predictor in virtually all studies. The following trivial example shows the presence of the distance effect when compatibility is held constant (hence, cannot explain the results): People decide faster that 49 is larger than 17 than they decide that 29 is faster than 17, when both pairs are compatible. The jury is still out on whether compatibility can serve as a decisive factor supporting analytic processing.

Another important point to note is that the compatibility effect does not rule out processing of the right-hand unit digit even when the decade suffices for a response. The notion of compatibility itself mandates processing of both the right and the left digit. Nevertheless, Verguts and De Moor (2005) and Brysbaert (1995) reported distance effects only when the two-digit numbers were confined to the same decade (for different decades, the [redundant] unit numbers apparently were not processed). Such decade breaks show that the unit digit is not always processed, providing evidence against an all-engulfing holistic model. Discontinuities found occasionally between the representations of one- and two-digit numbers seem also at odds with a strictly holistic model

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(Nuerk et al., 2001; Fias, Reynvoet, & Brysbaert, 2001; Ischebeck, 2003; see also Ganor-Stern et al., 2007; Ganor-Stern, Pinhas, & Tzelgov, 2009). According to Damian (2004), single digits are represented as pictures, but two-digit numbers are not, to wit "the . . . similarity between digits and pictorial stimuli probably breaks down outside the realm of single-digit numbers" (Damian, 2004, p. 170). The upshot is, the evidence for an analytic, componential model seems as strong (or as weak) as that for the alternative holistic model.

Deploying Multivariate Models of Perception by Way of a Resolution

The theoretical impasse invites deployment of advanced methods of analysis. These assays do not rely on magnitude comparisons (or other number tasks) as their point of departure, but rather treat multidigit numbers as a special case of multidimensional cognitive processing. The present effort included application of the Garner model (Garner, 1970, 1974b; Garner & Felfoldy, 1970 see Algom & Fitousi, 2016, for a recent review) and of the system factorial technology (SFT; Townsend & Nozawa, 1995; see Algom, Eidels, Hawkins, Jefferson, & Townsend, 2015, and Little, Altieri, Fific, & Yang, 2017, for recent reviews). We emphasize that these methods and attendant theory are of general appeal in cognition with applications running the gamut from face perception (Fitousi, 2015; Fitousi & Wenger, 2013; Wenger & Townsend, 2001) to other-race perception (Yang, Fifić, Chang, & Little, 2018) to attention under emotion (Mama, Ben-Haim, & Algom, 2013) to form and color perception (Fitousi, 2016, 2019; Garner, 1974a, 1974b) to categorization (Fific, Nosofsky, & Townsend, 2008; Little, Nosofsky, Donkin, & Denton, 2013). We pioneered to apply this pair of tools jointly to identify the properties of the system that governs perception of two-digit numbers.

Our novel exploration produced a number of important results. Perhaps the most meaningful new outcome was delineation of the processing characteristics of any given two-digit number. Virtually all modeling efforts to date are based on comparison of pairs of two-digit numbers. Our result, by contrast, refers to a single stimulus—to the two-digit number onto itself. In this sense, our result is more fundamental or primitive than results tailored to account for specific tasks (although our own model is also informed by existing data). A major thrust of our approach is exploration of the two-digit number as a stimulus object on a par with, say, a black car, a red apple, a human face, or a lemon tree. Contemporary cognitive science divides such objects—the objects of our perception—into those composed of *integral dimensions* (e.g., faces, colors) and those composed of *separable dimensions* (e.g., buildings of different shapes and colors, speed and the making of cars). The former objects are experienced as unified Gestalts, the latter as a combination of separately perceived components. Are two-digit numbers integral or separable? To decide, we subjected two-digit numbers to interrogation by the Garner paradigm and discovered that they are integral under some conditions, but separable under other conditions. Equipped with these insights, we were able to construe each two-digit number (not pair of numbers) as a Stroop stimulus, which sometimes did and sometimes did not generate the Stroop effect. Our investigation showed for the first time that the concept of "two-digit number" is an emergent property, which is formed late in processing.

Another equally important outcome of our investigation was resolution of the long-standing issue of parallel versus serial processing of two-digit numbers. This issue has been associated in the literature with the aforementioned issue of holistic versus decomposed representation. Now deciding between parallel versus serial processing is a daunting task (Townsend, 1990), with the distinction only tractable through dedicated rigorous formal means (Algom et al., 2015). A major problem is that of mimicry (e.g., Dzhafarov, 1997; Townsend & Ashby, 1983; Van Zandt, 2002). For example, virtually all parallel processing models can be mimicked by appropriate serial models (Algom et al., 2015; Van Zandt, 2002). Recall also that there exists a family of models within each mode of processing. For example, the benchmark parallel model entails independent parallel race of processing channels. Other parallel models, however, entail various types of cross-talk between channels (so that processing is still parallel but not independent). A parallel model with an extreme type of interaction is called the "coactive parallel model," where the different processing channels coalesce onto a single channel downstream. Yet another property distinguishing between models is the criterion adopted for emitting the response, a feature known as the "stopping rule" (Townsend & Nozawa, 1995). For example, in a serial type of processing the response can be given as soon as the first process is completed (minimum-time or self-terminating stopping rule) or after awaiting completion of the second (all) processing (exhaustive stopping rule). These delicate issues have not been considered in the literature on two-digit numbers. We do so here, taking full advantage of the state-of-the-art toolkit of the SFT.

Inspecting the literature, the vivid debate on processing started in earnest with the early (yet still one of the best) study by Poltrock and Schwartz (1984) who found a serial type of processing (with order guided by the place-value principle). The authors found that the time needed to compare two multidigit numbers "increased linearly with the position of the left-most pair of different digits (Poltrock & Schwartz, 1984, p. 44). This result was inconsistent with the results of a pair of closely preceding studies by Hinrichs and his associates (Hinrichs, Berie, & Mosell, 1982; Hinrichs et al., 1981) who found parallel holistic processing of two-digit numbers. Although Poltrock and Schwartz (1984) used some of the refined tools developed by Saul Sternberg on memory search (Sternberg, 1966, 1969), two decades later, Pixner, Moeller, Zuber, and Nuerk (2009) used the compatibility effect to adjudicate on the question of serial versus parallel processing. The authors argued that "parallel processing is indicated by the regular compatibility effect while sequential processing is associated with a negative compatibility effect (by which incompatible pairs are processed faster than compatible ones; Pixner et al., 2009, p. 42, note in brackets added). Concurrently, Moeller, Fischer, Nuerk, and Willmes (2009) attempted to decide the issue by using data from distributions of eye fixations. Subsequent more general models were still based on eye fixation and reaction time (RT) data (Huber, Cornelsen, Moeller, & Nuerk, 2015). This brief sample from a growing literature suffices to show that the tools used were not nearly optimal for evaluating the architecture underlying two-digit number processing.

To recap, discerning the architecture underlying two-digit number processing (serial vs. parallel) is absolutely paramount, but accomplishing the identification is an arduous undertaking. Nevertheless, it can and should be done (Townsend, 1990; Townsend

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& Ashby, 1983; see also, Williams, Eidels, & Townsend, 2014). This was a major goal of the present study.

The Garner Test: Are Two-Digit Numbers Integral or Separable?

Garner has put forward a fundamental partition between *integral* and *separable* dimensions that combine to create the objects in our perception (Garner, 1962, 1970, 1974a, 1974b; 1976; see also, Garner, 1991). This distinction is a pillar of today's cognitive science (Algom & Fitousi, 2016; Melara & Algom, 2003). It refers to the difference between stimulus objects that are experienced in their totality as undivided wholes (objects composed of integral dimensions) and stimulus objects that impress the observer merely as a composition of distinct attributes (objects composed of separable dimensions). Objects made of separable dimensions are readily decomposed into their constituent dimensions, whereas objects made of integral dimensions can be decomposed perceptually only imperfectly and with cognitive effort. The measure known as *Garner interference* provides a quantitative assay of the difference in processing between separable and integral dimensions. Concerning two-digit numbers, integral processing is consistent with a holistic representation, whereas separable processing is consistent with analytic representation. Are the decades and units composing two-digit numbers integral or separable? The Garner paradigm can be used to provide an answer.

There is an important caveat to issue at this point. To avoid circular reasoning, the results of the Garner paradigm, in particular the measure of Garner interference, cannot serve as the sole arbiter of the separability-integrality distinction. No investigator was more acutely aware of this constrain, and, in general, of the need for converging operations, than Garner himself (Garner, Hake, & Eriksen, 1956; see also, Garner, 1962, 1974b). In the absence of converging operation, any (theoretical) concept is simply a restatement of a particular experimental result. It is for this reason that the concept of "separable dimensions" is now sustained by converging operations from information theory (e.g., Garner & Morton, 1969; Fitousi, 2013), from similarity scaling (e.g., Attneave, 1950; Melara, 1992), and, indeed, from the SFT itself (Fific et al., 2008; see also, Algom, 2016 and Algom & Fitousi, 2016). Too, Ashby, Townsend, and their associates developed the edifice of general recognition theory to provide further independent nonempirically based definitions for the concept of separibility (and other related concepts; e.g., Ashby & Townsend, 1986; Silbert & Hawkins, 2016; Townsend, Houpt, & Silbert, 2012). The upshot is, the concept of separability (and integrality) is firmly established now as a principal guiding idea in cognitive science.

The Garner Paradigm

Consider our application of the paradigm to two-digit numbers. In three types of blocks, the observer decided, while timed, whether the left-hand decade digit was larger or smaller than five (while ignoring the unit digit). In the baseline block, the task-irrelevant right-hand unit digit was held at a constant range (e.g., always smaller than 5). In the filtering block, the task-irrelevant right-hand unit digit changed from trial-to-trial in a random fashion. In the correlation block, the task-irrelevant unit digit changed in tandem with the target decade digit (either both were larger than or both were smaller than 5 or one was always larger than 5 and the other always smaller than 5 or vice versa). Again, the task for the participant was the same in all the three blocks of trials: to classify the target digit of decade as speedily and accurately as possible.

The difference in RT between the baseline and the filtering blocks defines the measure known as *Garner interference* (Pomerantz, 1983):

$$\text{Garner Interference} = \text{MRT(Filtering)} - \text{MRT(Baseline)} \quad (1)$$

where MRT is the mean latency to classify the decade number (as greater or smaller than 5). An analogue formula exists for error. The presence of Garner interference—poorer performance in filtering than in baseline—shows that irrelevant variation in the unit number took a toll on performance with the decade number, compromising fully selective attention to the decades. Now, the complete Garner paradigm includes a third condition, correlation, where the irrelevant units also vary from trial to trial. However, under correlation they do so in a corresponding manner with the target dimension of decade. Under positive correlation, the decade and unit always match in magnitude (both large or both small), whereas under negative correlation, the decade and unit always mismatch in magnitude (one small, the other large). Performance under correlation is often better than that in baseline, the difference is called *redundancy gain*. The reason is obvious: The nominally irrelevant dimension of units now becomes predictive of the relevant dimension of decades (note that negative correlation is predictive of the target decade to the same extent as positive correlation). Note also that the presence of redundancy gain simultaneously signals the failure of fully selective attention to the target decades.

Formal Definition of Integral and Separable Dimensions

Equipped with these measures, Garner identified two general patterns of performance. Separable dimensions exhibit uniform performance, so that speed (and accuracy) remains invariant across baseline, filtering, and correlation. The participants neither suffer Garner interference (in filtering) nor reap redundancy gain (in correlation). For integral dimensions, performance is best under correlation and worst in filtering. With integral dimensions, the participants suffer Garner interference from orthogonal variation in filtering but reap redundancy gain from corresponding variation in correlation.

Therefore, the following RT equalities define separable dimensions:

$$\text{MRT(correlation)} = \text{MRT(baseline)} = \text{MRT(filtering)} \quad (2)$$

where MRT is the mean RT to respond to the relevant dimension. By contrast, integral dimensions are defined by the following RT inequalities:

$$\text{MRT(correlation)} < \text{MRT(baseline)} < \text{MRT(filtering)} \quad (3)$$

where MRT is again the mean RT to respond to the target dimension. In Figure 1, we illustrate the Garner paradigm along with the two general patterns of results.

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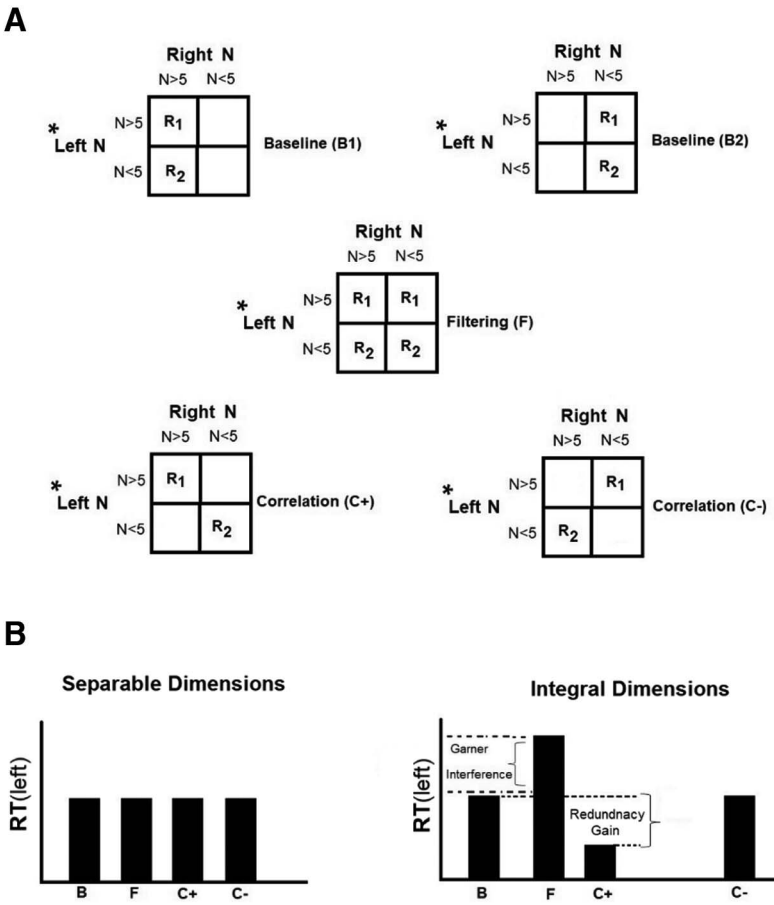


Figure 1. Schematics of Garner's speeded classification paradigm used in the study. The three basic conditions—baseline (B), filtering (F), and correlation (C)—are depicted on the top. The asterisk indicates that the left number (N) standing for the decade, not the right number standing for the unit, was the relevant one for responding. R1 and R2 depict the correct responses to the two ranges of values of the relevant dimension, left N. The correlation conditions depict two possible ways of correspondence in the appearance over trials of the left and right number. Two prototypical outcomes, tapping integral and separable processing, are depicted at the bottom.

Asymmetric Patterns

We administered the same three types of blocks of trials once again, but now with the units as the target dimension. The resulting pattern is identified by the same means. At this point, we issue an important caveat: Given a pair of dimensions, A and D, A can be separable with respect to D although D is integral with respect to A (Garner, 1976; Melara & Algom, 2003; Melara & Mounts, 1993; Sabri, Melara, & Algom, 2001). The pattern of interference need not be symmetric across judgments of the two dimensions of the same object. A ready example is provided by the well-known Stroop effect: Although ink color is integral with respect to the carrier word (hence the Stroop effect), the word is separable with respect to ink color (e.g., Eidels, Townsend, & Algom, 2010; Melara & Marks, 1990; Melara & Mounts, 1993; see Pansky & Algom, 1999, 2002, for asymmetric patterns with numbers). Concerning two-digit numbers, it is possible that people neither suffer Garner interference nor reap redundancy gain when judging the decades, but that they do both when judging the units. One must keep in mind that separability or integrality of a given object is contingent on the tested dimension.

In sum, identification of the Garner pattern(s) provides powerful support for the pertinent representation by falsifying classes of candidate models. Before discussing the predictions in detail, we first present our second tool, the SFT.

SFT Interrogation of Two-Digit Numbers: Is Processing Parallel Independent or Interactive?

The SFT is a theory, sustained by an advanced methodological toolkit, aimed at uncovering the mental infrastructure of the cognitive system under test. To unearth the mental infrastructure, the following four (distinct yet interrelated) features are specified within SFT: (a) architecture (serial vs. parallel, and, if the latter, coactive or not); (b) stopping rule (self-terminating vs. exhaustive); (c) capacity (limited vs. unlimited vs. super), and (d) stochastic independence (independence vs. dependence). We discuss each of these characteristic and the theorems relating them below,

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but the reader is also advised to consult recent extensive tutorials entailing more detail and mathematical rigor than we can manage here (see in particular the edited volume by Little et al., 2017). For applications similar to the present one, the reader might wish to consult Fitousi and Algom (2018) and Algom, Fitousi, and Eidels (2017).

First consider the current SFT task. Presented with a pair of numerals, the observer detected the presence of at least one target—a numeral greater than 5—regardless of its place in the pair (left, right, or both). The observer responded yes if the decade number (ND) was large (i.e., ND $>$ 5), if the unit number (NU) was large (NU $>$ 5), or if both were large (i.e., ND $>$ 5 *and* NU $>$ 5). Note that target detection in this case mandates divided attention due to the nonselective nature of the task. Next consider the experimental setup, which entailed a pair of factorial designs. The first one (factorial design A in Figure 2) was created by crossing the numerical magnitude of the decade number (large, small) with that of the unit number (large, small). The second factorial design (B in Figure 2) was based on the added manipulation of perceptual salience. Although this manipulation was done throughout all presentations, considering only the double-target trials at the upper left of Figure 2 (i.e., trials in which both the decade number and the unit number are larger than 5) produces a second factorial design. It is created by crossing the salience (high, low) of the decade target with the salience (high, low) of the unit target. This version of the SFT experiment is known as the *double factorial paradigm* (DFP; Townsend & Nozawa, 1995; see also, Eidels et al., 2010; Fitousi & Wenger, 2013; Algom et al., 2015). We illustrate our design in Figure 2.

The data generated by the pair of factorial designs entailed in the DFP are then analyzed by the SFT network of theorems and attendant RT inequalities in order to pinpoint the underlying mental architecture. These involved analyses are needed because the mean RTs (used in existing literature) cannot differentiate among candidate models. For example, serial and parallel systems can mimic one another (exhibiting the same pattern of RTs), hence one needs the concepts of selective influence and mean interaction contrast (MIC) to address for resolution (Schweickert & Townsend, 1989; Townsend, 1984; see also Eidels, Houpt, Altieri, Pei, & Townsend, 2011).

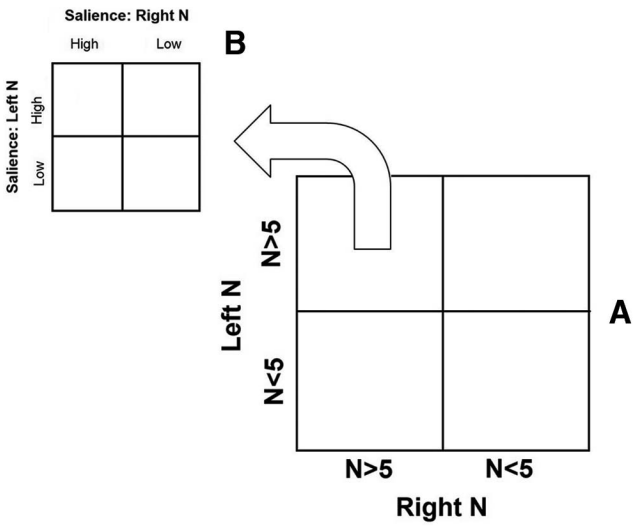


Figure 2. Schematics of the double factorial paradigm used in the study. Each member of the two-digit number is denoted by N. With N $>$ 5 defined as target, the factorial design at the bottom (factorial design A) creates double-target (upper left), single-target (cells on the minor diagonal), and no-target (lower right) stimulus displays. The concurrent manipulation of salience creates a second factorial design (factorial design B) confined to the subset of double-target presentations (upper left).

Let us now consider several plausible models. The decade and the unit signals can be processed sequentially, so that processing does not overlap in time. In such serial processing, the response can be given upon completion of the first process (minimum time rule) or only upon completion of the second process (exhaustive rule). Alternatively, the two stimuli can be processed simultaneously, that is, in a parallel manner. In the parallel mode, the stimuli can be processed through independent channels so that the activation in one channel does not affect the activation level in the other channel. In other parallel models, the channels communicate with one another so that one channel facilitates (or inhibits) processing in the other channel. Most parallel architectures are further qualified by the operative stopping rule. Schematics of these standard models is presented in Figure 3.

How can one decide the veridical model? The key data for a decision are those included in the second factorial design (design B) at the upper left of Figure 2. The particular factorial plot resulting from that design provides diagnostic information regarding the candidate model. The concepts of *MIC* and *survivor interaction contrast* (*SIC*) that we discuss next refer to this factorial design.

Model Diagnosis: The MIC

Five possible factorial plots, based on the mean RTs in each cell of factorial design B, are presented at the left column of Figure 4. The plots are selectively associated with the models depicted in Figure 3. Now a simple algebraic routine can be used to represent each factorial plot by a single number. That number is called the MIC. The routine is as follows. Take each line in the plot in turn and calculate the (vertical) difference between the line's endpoints. Then, calculate the difference of the differences. This double-difference is the MIC:

$$\begin{aligned} \mathrm{MIC} &= [\mathrm{MRT}_{\mathrm{LL}}(t) - \mathrm{MRT}_{\mathrm{LH}}(t)] - [\mathrm{MRT}_{\mathrm{HL}}(t) - \mathrm{MRT}_{\mathrm{HH}}(t)] \\ &= \mathrm{MRT}_{\mathrm{LL}}(t) - \mathrm{MRT}_{\mathrm{LH}}(t) - \mathrm{MRT}_{\mathrm{HL}}(t) + \mathrm{MRT}_{\mathrm{HH}}(t), \end{aligned}$$

where L and H stand for low salience and high salience, respectively.

Let us consider the meaning of each pattern in Figure 4 with respect to two-digit number processing. If the processing of the left and the right digits is performed in a strictly serial fashion, the MIC equals zero regardless of the operative stopping rule (self-terminating or exhaustive). In contrast, parallel self-terminating processing and parallel coactive processing both yield an overadditive pattern (MIC $>$ 0). The reason is that one expects fast responding when either or both dimensions entail a salient target. The single combination, LL, yielding slower responses, is what produces the overadditive pattern. Finally, parallel exhaustive processing results in an underadditive pattern (MIC $<$ 0). Under this regime, RT is determined by the slower of the two dimensions,

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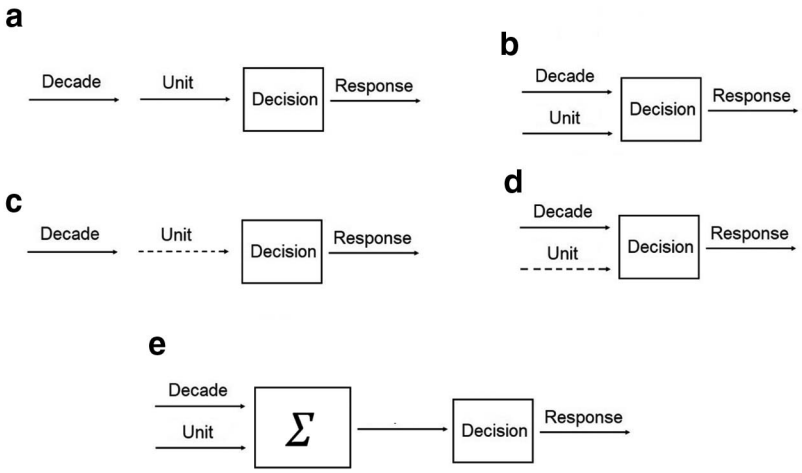


Figure 3. Illustrations of serial (a, c), parallel (b, d) and coactive (e) models. Systems a and b depict an exhaustive stopping rule, where both processes must finish before a decision and response is made. Systems c and d depict a minimum-time stopping rule, where processing ceases once the decade process is completed (but the units process had not finished, as indicated by the broken lines). Finally, system e illustrates a coactive mode of processing where activation from the two channels is summed up before decision. Note that in the illustrations of serial models, the decade process preceded the unit process, but the reverse order is also possible. Adapted from Figure 4.5 in "Features of Response Times: Identification of Cognitive Mechanisms Through Mathematical Modeling," by D. Algom, A. Eidels, R. X. Hawkins, B. Jefferson, & J. T. Townsend, 2015, in J. R. Busemeyer, Z. Wang, J. T. Townsend, & A. Eidels (Eds.), *The Oxford Handbook of Computational and Mathematical Psychology*. Oxford, UK: Oxford University Press. Copyright by the Oxford University Press.

which means that performance will be fast only for the HH combination. These intuitions are supported by formal proofs (Townsend & Nozawa, 1995). The main results are summarized in Table 1.

Although the MIC is a valuable statistic, it is not fully diagnostic of the processing model. As can be seen in Table 1, the MIC does not distinguish between serial exhaustive and serial self-terminating processing as well as between parallel self-terminating processing and parallel coactive processing. However, diagnostic power is increased when one considers the entire distribution rather than the means. A powerful such measure is the SIC.

Model Diagnosis: The SIC

Derivation of the SIC is straightforward. Subdivide the entire RT range into small time-bins and calculate in each bin the MIC. The aggregation then yields the SIC(t) function with the proviso that the calculation within each bin is made with respect to the values of the pertinent survivor functions (rather than with the means). The survivor function, S, for a random variable, T (corresponding to time of processing), is the probability that T takes longer to complete than t, so that $S(t) = p(T > t)$. Note that $S(t) = 1 - F(t)$, with $F(t)$ denoting the distribution function for the same process. To recap then, in a manner analogous to the calculation of the factorial plot with respect to the mean RTs, one calculates in each time bin, t, the values of the survivor function associated with each of the four types of target combinations, $S_{LL}(t)$, $S_{LH}(t)$, $S_{HL}(t)$, and $S_{HH}(t)$. The SIC is given by:

$$SIC(t) = [S_{LL}(t) - S_{LH}(t)] - [S_{HL}(t) - S_{HH}(t)]$$
$$= S_{LL}(t) - S_{LH}(t) - S_{HL}(t) + S_{HH}(t)$$

The SIC(t) functions for each model are presented at the right of Figure 4. Considering them in tandem with Table 1 yields distinctive predictions with respect to serial processing as well as with respect to parallel self-terminating and parallel coactive processing. The SIC(t) function yields fine-grain information, although the fact that the MIC is actually the integral of the SIC(t) puts constrains on the possible distribution of areas under the SIC(t) curve. For serial self-terminating processing, the SIC(t) is equal to zero at all time, t. In contrast, for serial exhaustive processing, the SIC(t) is negative at early values of t, but positive at latter values. Notably, the total area spanned by the negative portion should equal that by the positive portion. The two species of parallel processing are also distinct now. The SIC(t) is positive at all values t for self-terminating parallel processing, but it changes from an early moderate negativity to a predominant positivity at later values of t. Again, these SIC(t) results are supported by dedicated theorems (Townsend & Nozawa, 1995; see also, Little et al., 2017). We conclude our synopsis by discussing the SFT measures referring to the basic factorial design (design A) in Figure 2.

Capacity-Based Diagnosis: Sensitivity of the System to Changes in Workload

The developments discussed thus far referred to the data emanating from the factorial design at the upper left of Figure 2 (design B), the subset of trials including targets only. We now discuss the data included in factorial design A of Figure 2. The data from factorial design A is best suited to decide the capacity of the system processing two-digit numbers. Workload capacity or capacity refers to changes in performance when load—the number of targets to processes—is varied. As we shall see below, if the rate

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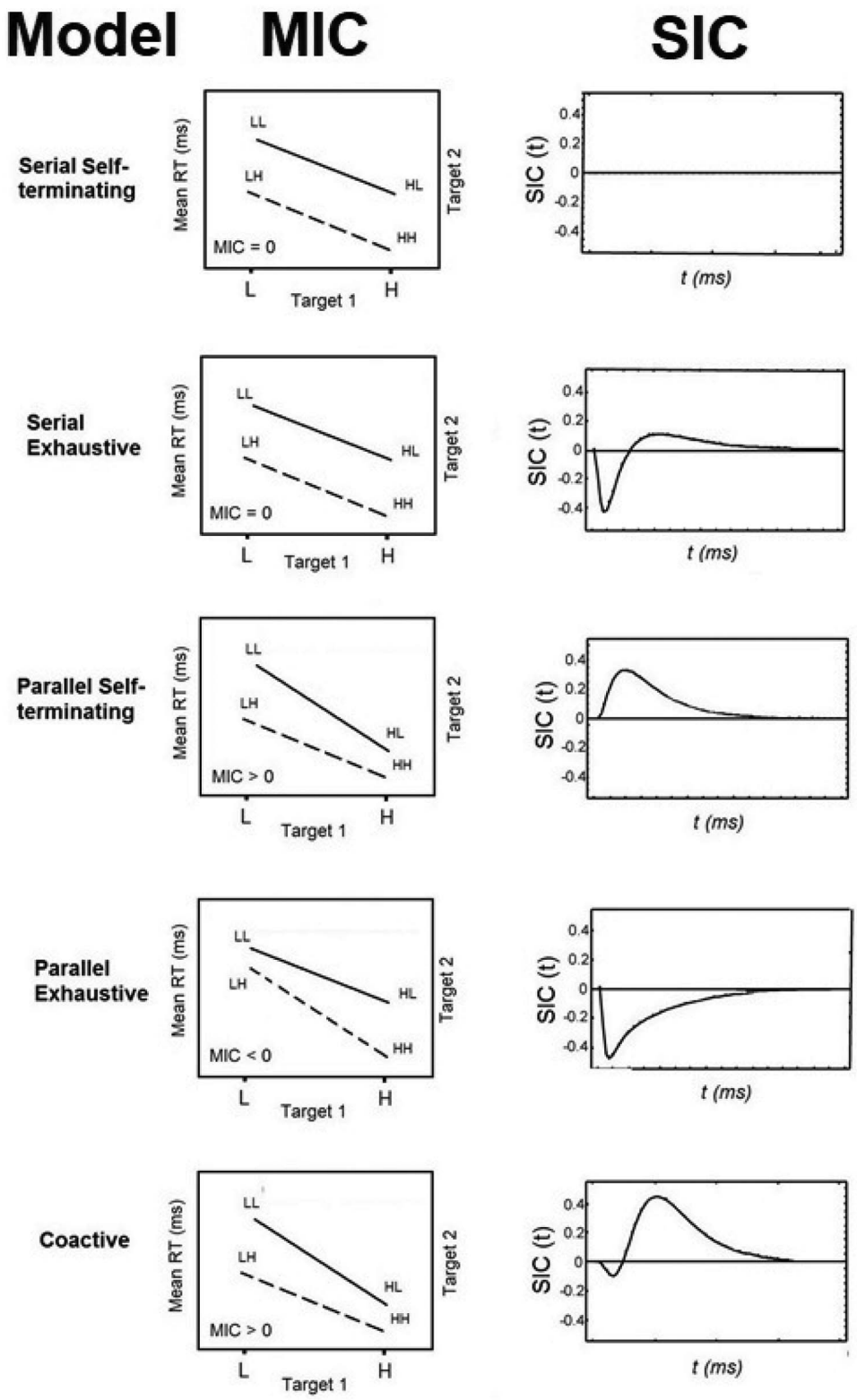


Figure 4. The model (left panel), the associated factorial plot with the mean interaction contrast (MIC; middle panel), and the associated survivor interaction contrast (SIC) functions (right panel) for the five models depicted in Figure 3. See text for the meaning and definition of SIC.

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Table 1
Mean Interaction Contrast (MIC) and Survivor Interaction Contrast (SIC) Predictions for Reaction Times in Five Processing Models

Architecture	Stopping rule	Predictions
Parallel	Minimum time	MIC $>$ 0; SIC(t) $>$ 0
Parallel	Exhaustive	MIC $<$ 0; SIC(t) $<$ 0
Serial	Minimum time	MIC $=$ 0; SIC(t) $=$ 0
Serial	Exhaustive	MIC $=$ 0; SIC(t) $<$ 0 for small t, $>$ 0 for large t
Coactivation		MIC $>$ 0; SIC(t) $<$ 0 for small t, $>$ for large t

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of processing in one channel remains invariant when another target is added, the system is of unlimited capacity. If presenting an additional target slows down processing in the first channel, then capacity is limited. If presenting that additional target speeds up processing in the first channel, then capacity is super. Note that factorial design A entails variation in the number of targets: two (at the upper left) and one (along the minor diagonal). Is detection performance (saying YES to the presence of at least one target) superior with two targets present than with only one target present? If so, there is a redundant target effect (RTE). Does the presence of an RTE indicate an interaction or dependence in processing of the two targets? This question was first pursued via an RT inequality suggested by Miller (1978, 1982), known as the race model inequality,

$$F_{\mathrm{DU}}(t) \leq F_{\mathrm{D}}(t) + F_{\mathrm{U}}(t),$$

in which $F(t)$ is, again, the distribution function for decade-target only (D), unit-target only (U) the double-target (DU) presentations. According to the inequality, the distribution function for the double-target trials cannot exceed the sum of the single-target distribution functions—if indeed the decades and units are independent in processing (i.e., if there is a parallel race without cross-talk between the two channels). Conversely, violations of the inequality have been interpreted to reject parallel race in favor of coactivation.

However, an RTE can still emerge when the race model inequality is *not* violated, although the magnitude of the double-target advantage can be fairly limited (Biederman & Checkosky, 1970; Raab, 1962). The Miller inequality is recast in SFT as a statement about the capacity of the system. Again, if the rate of processing a target is slowed down when presented along with another target, the system is *limited capacity* (i.e., there is not an RTE). When processing a target is the same in single and in joint presentations, the system is of *unlimited capacity*. Notably, if the presence of two targets speeds up the ability of the system to process each target, the system is *supercapacity*. Townsend and Nozawa (1995) proved that when the Miller inequality is violated, the system is supercapacity (at that t). The capacity of the observer's system is captured formally by the capacity coefficient, $C(t)$,

$$C(t) = \frac{H_{DU}(t)}{H_D(t) + H_U(t)}$$

where $H(t)$ is the integrated hazard function, h,

$$H(t) = \int_0^\infty h(t)dt = -\log[S(t)]$$

When $C(t) < 1$, the system is limited capacity, whereas $C(t) = 1$ attests to unlimited capacity. Supercapacity is indicated by $C(t) > 1$. Although logically distinct from the other features of the system, capacity naturally allies with some properties more than with others. For example, the presence of supercapacity is consistent (although not necessary) with coactivation, but not with independent race.

Finally, whereas the Miller inequality sets an upper bound on RT with double-target displays, Grice and his colleagues (Grice, Canham, & Boroughs, 1984) suggested a lower bound,

$$F_{DU}(t) \geq MAX[F_D(t), F_U(t)].$$

If the Grice bound is violated, then the system is of limited capacity to a rather serious degree (certainly, any RTE will be absent). In that case, the increase in workload from one target to two by targets strains (rather than augments) performance. This concludes our presentation of the main features of SFT. We next turn to the predictions that can be made on the basis of the joint application of the Garner and the SFT technologies.

Predictions Based on the Combined Examination of Two-Digit Numbers

The Garner and the SFT tasks complement and constrain each other, hence our double-pronged attack permits strong converging of evidence (Garner, Hake, & Eriksen, 1956). First, consider the overall complementarity: Whereas the Garner battery is psychology's litmus test of separability and integrality, the detailed time course information entailed in the SFT enables one to select among the candidate processing systems conductive to the observed separability or integrality. This was our tactic here. Second, note the difference in task. Garner's is that of *selective attention*: The observer focuses exclusively on the target dimension (decade or unit) while ignoring the other, task-irrelevant dimension. The SFT task is one of *divided attention*: The observer attends to both dimensions to accomplish the task of nonselective target detection. Third, there's the issue of (a)symmetry: The Garner outcome is dimension-dependent. The pattern can be symmetric or asymmetric across the two dimensions, for example, separable for decades (when judging the left digit), but integral for units (when judging the right digit). In contrast, the SFT is blind to the place of entry to the two-digit number. The SFT outcome is a unitary pattern pertaining to the two-dimensions as a whole. This difference informs the combined analysis and guides the predictions.

If the Garner outcome is symmetric-separable, namely, a separable pattern is found for both the decades and the units then

independent processing models are supported. Simultaneously, interactive SFT models, including notably coactivation, are challenged. Our task would be then to tap the underlying model among species of sequential processing and parallel independent processing. Conversely, unearthing an integral pattern for both decades and units would support coactivation and other species of interactive processing, but falsify independent-processing models. The SFT then serves to identify the specific interaction that generated the integrality. However, the truly challenging situation arises when the Garner outcome is asymmetric. This means that either the decades or the units are integral in processing. Because the SFT is blind to the place of entry, any Garnerian integrality, whether decades or units, supports an interactive processing model of SFT.

Experiment 1: Diagnosis of Separability and Integrality by Garner Interference

Method

Participants. Twenty undergraduate students from Ariel University participated (all women, *M* age = 21.9, *SD* = 1.68). They received course credit for their participation. All reported normal or corrected-to-normal vision. The experiment (as well as Experiment 2) received the approval of the Ariel University Ethics Committee, and all participants provided their agreement to perform in the experiment.

Stimuli. The stimuli were all two-digit numbers from 11 to 99 that did not include 5 or 0 as one of their digits, making for 64 stimuli in all (see Appendix A). The numbers were presented in black over the gray background of the monitor at a center position (Times New Roman, font size 18). These stimuli extended 0.8 cm in height and 1.1 cm in width and were viewed from a distance of 60 cm. Regarding location, we introduced a trial-to-trial visual uncertainty of up to 10 pixels at around the fixation point (within the standard 640 × 480 display).

Design. The experiment included two tasks: judging the decade number or the unit number. Each task included five blocks of trials separated by short breaks (two baseline blocks, a filtering block, and two correlation blocks). The five blocks of a task were presented in one sequence (with breaks in-between), and only then was the other task presented (again with five blocks of its own). Again, the tasks were defined by the relevant dimension for responding: decade or unit.

The decade task. The participant decided on each trial whether the decade digit was greater or smaller than 5 (while ignoring the unit digit). There were two baseline blocks with the task-irrelevant unit held at a constant range: In one block, all unit digits were smaller than 5, whereas in the other block all unit digits were larger than 5. In the filtering block, the digits of both dimensions varied in an orthogonal fashion so that all the two-digit numbers were presented in a random order. Finally, there were two correlation blocks with the task-irrelevant unit varying from-trial-to-trial in a corresponding fashion with the target decade digit. In one block, the values of the two dimensions varied in tandem such that the decade and unit digits both were greater than 5 or both were smaller than 5 (C+, or positive correlation). In the other block, a greater than 5 decade digit was always paired with a smaller than 5 unit digit, and a smaller than 5 decade digit was always paired with a greater than 5 unit digit (C-, negative correlation).

The unit task. The same five blocks were presented, but now with the unit digit as the relevant dimension for responding. The participant decided on each trial whether the unit digit was greater or smaller than 5 (while ignoring the decade digit). See Appendix A for the list of numbers included in each block in the two tasks.

Two features of the design are noteworthy. First, there is a slight departure from the original Garner design, which includes only two stimuli per dimension. Here, a range of values (numerals greater than 5 and numerals smaller than 5) stand for each stimulus value (see Ganel & Goshen-Gottstein, 2002, for a similar multivalue study, and see the review by Algom & Fitousi, 2016). This slight difference in no way affects the derivation and interpretation of all Garner effects. Second, given that all stimuli in the positive correlation condition are decade-unit congruent and that all stimuli in the negative correlation condition are incongruent, it is possible to derive a Stroop effect across these conditions. Of course, the stimuli in filtering (and even at baseline) permit derivation of the usual within-condition Stroop effect.

Procedure. The participant first completed the five blocks of one task (e.g., unit), then performed in the five blocks of the other task (decade). Each of the two baseline blocks consisted of 32 trials as did each of the correlation blocks. The filtering block included 64 trials. Therefore, the participant performed in 192 trials per task in all. The order of tasks was random as was the order of blocks within a task, the latter subject to the proviso that the two baseline blocks and the two Correlation blocks were completed in a sequence. Brief breaks separated each of the blocks within a task and a somewhat longer break separated between the tasks.

Each block started with a blank screen for 500 ms, followed by the presentation of a two-digit number at the center of the screen (subject to spatial uncertainty of 10 pixels). Stimulus exposure was response terminated, with an intertrial interval of 350 ms (between response and the following stimulus). Participants responded manually. They responded with a right-hand key ("m") if the number was larger than 5, and with a left-hand key ("z") if the number was smaller than 5. This response mapping was constant across all participants.

Data analysis. Trials with RTs shorter than 150 ms or RTs longer than 2500 ms were excluded from analysis (0.7% of all trials). Trials with incorrect responses were also removed (7.2% of all trials).

Results

First consider the RT data which were subjected to a three-way analysis of variance (ANOVA) with task (decade, unit), block (baseline, filtering),[2] and congruity (decade-unit congruent, decade unit incongruent) as factors. The only significant effects were the two-way interactions of Task × Block, $F(1, 19) = 5.52$, $MSE =$

[2] We did not include the correlation blocks in this analysis for two reasons. First, under correlation congruity is separated into pure blocks of congruent only and incongruent only stimuli, thereby altering its typical role in the other blocks and in the Stroop literature at large. Second, the correlation condition has no bearing on the measure of Garner interference, which is the focus of this analysis. We subsequently discuss in detail the results obtained in the correlation condition.

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12,803, $p < .05$, and of Task × Congruity, $F(1, 19) = 4.95$, $MSE = 12{,}803$, $p = .05$. The first interaction pertains to Garner effects, the second to Stroop effects. We first discuss the Garner results. The presence of the interaction means that any Garner interference in the data was modulated by task. Planned comparisons indeed revealed that a large Garner interference of 55 ms plagued performance in the unit task, $t(39) = 1.74$, $p < .05$, but none emerged in the decade task—filtering was slower by mere 1 ms than baseline; $t(39) = 0.19$, $p = .60$. The BF value of 0.23 in a Bayesian analysis provided strong evidence against the presence of Garner interference in judgments of the decade digit. Our participants attended to the target decade digit without suffering distractions from orthogonal variation in the task-irrelevant unit digit, but they could not attend to the unit digit without being affected by the decade digit. The lack of a main effect for task ($F < 1$) is also notable; as shown in Table 2, which summarizes the Garner data, average Baseline performance was 602 and 597 ms, respectively, for decade and unit classifications. The parity is not surprising given that the participants were judging single digits in both tasks. At the same time, the parity rules out explanations that attribute decade-unit asymmetry in performance to mismatched difficulty of the two dimensions at baseline. As can be also seen in Table 2, the error rates largely followed the pattern of the RTs, but the only significant effects in the overall ANOVA were those of block, $F(1, 19) = 5.68$, $MSE = 0.022$, $p < .05$, and Task, $F(1, 19) = 4.54$, $MSE = 0.016$, $p < .05$.

In the language of Garner, selective attention was perfect for the left-hand decade digit, but it failed for the right-hand unit digit. Garner interference was absent with the former, but it was robust with the latter. Applying this benchmark measure of dimensional interaction, two-digit numbers comprise separable dimensions with their decade as the relevant dimension, but they are integral dimensions with units as the relevant dimension.

The above conclusions should be qualified (but not rejected) when considering the results in the correlation blocks, but let us first return to the Stroop results in the overall ANOVA. The presence of the Task × Congruity interaction indicated that any Stroop effect in the data was modulated by task. Planned comparison revealed that Stroop effects were present in the unit task, $t(39) = 1.88$, $p < .05$, but not in the decade task, $t(39) = 0.02$, $p = .49$. Consequently, we considered the Stroop effects (for RT and error) separately for the two tasks in the Baseline and the Filtering blocks. Again, Stroop congruity was defined as the difference between RT to incongruent (decade-unit mismatching) and congruent (decade-unit matching) stimuli in a task. A positive difference thus indicates shorter RTs to congruent stimuli, that is, the presence of a Stroop effect. For each participant we derived these Stroop scores at baseline and in filtering, and the means appear in Table 3.

Table 2
Mean Reaction Times (RTs; in Ms) and Proportions of Error (%) for Judgments of Units and Decades in the Baseline, Filtering, and Correlation Conditions Dimension

	Units				Decades			
	RT		Error (%)		RT		Error (%)	
Error (%)	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Baseline	597	25	10.5	3.3	602	24	6.0	1.5
Filtering	652	26	9.9	3.2	603	25	5.5	1.3
Positive correlation	611	22	6.4	2.4	574	21	4.5	.8
Negative correlation	620	27	12.8	4.1	635	29	9.4	3.4

Again, a striking feature of the data shown in Table 3 is the complete absence of Stroop congruity for decades, but their presence in force for units. For the former, the momentary value of the irrelevant unit digit, congruent or incongruent with the target decade digit, did not make any difference. For the latter, congruent stimuli in which the irrelevant decade digit matched the relevant unit digit, were judged faster that incongruent stimuli in which the two digits mismatched. The error data closely mimicked those of the RT. These results with the Stroop effect mirror the asymmetry observed with the Garner results. Both measures show that the processing of decades was immune to intrusions from irrelevant unit, but that the processing of unit was strongly affected by irrelevant decade.

It is worth pausing a bit to understand the Stroop-Garner relation: Whereas Garner interference shows that the observer noticed irrelevant variation, the Stroop effect shows further that the observer noticed also the kind of variation (matching or mismatching with the target). Because the observer cannot experience "kind of variation" without also experiencing "variation," Stroop congruity implies Garner interference although the reverse does not hold. Now this *logical* relation often holds (e.g., Melara & Algom, 2003; Melara & Mounts, 1993; Pomerantz, 1983, 1986; Pomerantz, Pristach, & Carson, 1989), but also often fails to hold in empirical data (e.g., Eitan, Schupak, Gotler, & Marks, 2014; Martino & Marks, 2000; Patching & Quinlan, 2002; van Leeuwen & Bakker, 1995). Although both the Stroop and the Garner variety assess (the failure of) selective attention, they do so along distinct and completely independent routes with the Garner assay the more fundamental and general tool (see Algom, Chajut, & Lev, 2004, for a detailed discussion). The cardinal distinction between integral and separable dimensions is based on the wealth of measures included in the Garner toolkit, and further Garnerian ideas and nomenclature are shared by routines of stochastic modeling like the general recognition theory and the SFT. The Stroop effect, when it applies, can complement the Garner results. Incidentally, the absence of a Task × Block × Congruity effect, $F(1, 19) = 0.63$, $MSE = 5214$, $p = .44$, attested to the independence of the Garner and Stroop effects in the current data, too.

Next consider the results in the correlation blocks in Table 2. The large Stroop effect observed for the decade task (as well as the large redundancy gains and losses) threatens our conclusion that two-digit numbers are separable when processed along their decade number. The difference between the blocks of positive and negative correlation when judging decade amounted to a highly significant Stroop effect of 61 ms, $t(18) = 3.71$, $p < .005$, $d = 0.48$. This result is perplexing because it implies that decades and units interact in the processing of decades, an interaction that is conspicuously absent in other portions of the data. For the decade digit, the absence of Garner interference and Stroop congruity in baseline and filtering implies separable processing, whereas the presence of Stroop congruity and redundancy gain in correlated contexts suggests integral processing.

To resolve this unique pattern of data (but see Algom et al., 1996, for a similar pattern), first note that the correlation task

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Table 3
Decade-Unit Stroop Effects: Mean Reaction Times (RTs; Ms) to Congruent and Incongruent Stimuli in the Decade and the Unit Tasks in the Baseline and Filtering Conditions

	Congruent		Incongruent		Stroop congruity	
Task	RT	Error (%)	RT	Error (%)	RT	Error (%)
Decade						
Baseline	602	5.0	597	7.2	−5	2.2
Filtering	595	4.4	595	6.2	0	1.8
Unit						
Baseline	582	5.8	635	15.4	53	9.6
Filtering	628	5.3	648	15.1	20	9.8

provides for a weak test of separability (e.g., Ashby & Gott, 1988; Ashby & Maddox, 1990; Ashby & Townsend, 1986; Maddox & Ashby, 1996). Under correlation, the observer can use an integrative rule even for separable dimensions to maximize performance (so that it is optimal for the observer to attend to the nominally irrelevant dimension due to its predictive power). Possibly, our observers ignored the irrelevant dimension in orthogonal contexts (to avoid interference) but paid attention to it in correlated contexts (to reap gain). However, this explanation still ignores the difference in performance between positive and negative correlation (with binary-valued dimensions, the two are equally predictive of the target dimension). The missing ingredient is the difference in semantic processing across filtering and correlation (Algom et al., 1996; see also, Algom, 2016). To enable separability in filtering, the observers did not process the irrelevant units semantically—qua numbers—but, possibly, as shapes. By contrast, the observers afforded full semantic processing to the units—processed them as numbers—in correlated contexts, the worse performance under negative correlation deriving from the semantic clash in this condition. The upshot is, the dimensions of two-digit numbers are separable when considered by their decades.

A final feature lurking in the data is the distance effect, which is also suggestive of a distance-unit asymmetry in processing. As is evident in Figure 5, a distance effect emerged for judgments of both the decade digit ($t = -5.17$, $p = .03$, $R^2 = 0.93$) and the unit digit ($t = -4.44$, $p = .04$, $R^2 = 0.90$). However, as is also evident in Figure 5, the effect was a bit larger and more pronounced for the decade digits ($t = 2.4$, $p = .059$). For the left (decade) digit, the participants responded to stimuli separated by a distance of four digits from the boundary of 5 over 100 ms faster than to stimuli merely 1 digit apart from 5; for the right (unit) digit, by contrast, the same difference was 74 ms. The difference in magnitude of the distance effect is notable given that exactly the same instructions ("larger or smaller than 5") and stimuli (1–9) were presented for both dimensions. Our participants were a bit more sensitive to the relative magnitude of the same digits in a left position that in a right position.

In sum, the results of Experiment 1 tell a coherent story. The processing of the components of two-digit numbers is asymmetric, favoring the left-hand number. When attending to the left-hand number, the processing is efficient and immune to intrusions from the task-irrelevant right-hand number. In contrast, people cannot attend exclusively to the right-hand number without concurrently noticing the task-irrelevant left-hand number. Whence the asymmetry? It likely derives from people's deep-seated knowledge of the place-value system, acquired through years of education. Because the leftmost digit is the most significant one in any multidigit number, our participants attended to the decade digit even when irrelevant to the task. The observers in the Garner task were forced to break the structure of the two-digit number by focusing in turn at each component (as though each was a single digit), yet they still preserved the syntax of two-digit numbers. The success or failure of selective attention to the component digits depended on the top-down knowledge of the place-value principle, even when the task did not require a representation of the number's global magnitude.

Given the asymmetric Garner pattern, in particular the integrality observed for the units, completely independent processing models are ruled out. The following SFT experiment and analysis thus aimed to identify the appropriate interactive model.

Experiment 2: Diagnosing the Processing Model Supporting Decade-Unit Interaction

Method

Participants. Four undergraduate students from Ariel University (M age = 23, SD = 1.82, two women) who did not take part in Experiment 1 performed in Experiment 2. All had normal or corrected-to-normal vision.

Stimuli. The stimuli were those of Experiment 1: all two-digit numbers between 11 and 99, excluding those with the digits 5 or 0. All the details of presentation and timing were also the same as in Experiment 1, but for a single notable exception. In Experiment

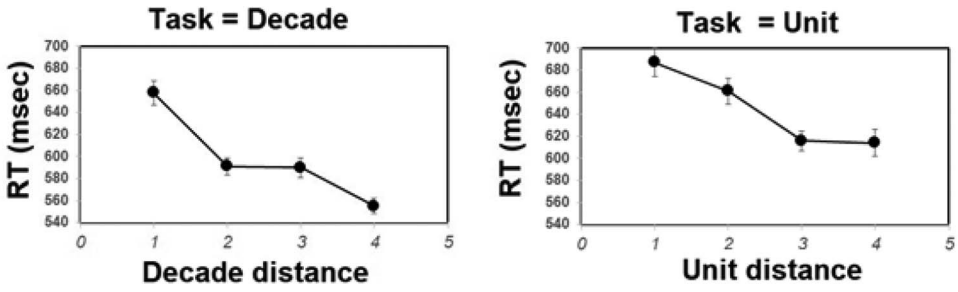


Figure 5. Distance effect for the decades digit in the decade task (left panel), and for the unit digit in the unit task (right panel).

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2, each component digit appeared in a highly salient and easily legible font and also in a less salient and legible font. The manipulation of salience was done by increasing the brightness of the digit appearing over the black background of the screen. The SFT task was that of speeded target detection. Targets were defined as any digit (decade or unit) that is greater than 5. Of the 64 two-digit numbers, 32 included one target, 16 included two targets, and 16 included no target (see Appendix B for detailed lists).

Design. All stimuli included in the two factorial designs of Figure 2 were presented, and these presentations were repeated in seven blocks of trials. Each block consisted of 384 trials with an equal number of "target absent" and "target present" trials. The latter distributed evenly across the three possible positive conditions (i.e., 64 double-target trials, 64 decade-target-only trials, and 64 unit-target-only trials). Each separate two-digit number repeated exactly the same number of times (12), with each appearing exactly three times in each of the four possible salience combinations (decade-salient/unit-salient, decade-salient/unit-faint, decade-faint/unit-salient, and decade-faint/unit-faint). Thus, the probability of each two-digit number in each salience condition was the same. This was also true for the stimuli in the "target absent" condition. Each observer performed in 2 consecutive days of testing, completing seven blocks of 384 trials (= 2688 trials) per day. These amounted to a total of 5,376 trials per observer.

Procedure. The participants were tested individually in a dark and a silent room. They were asked to give a YES response if the decade digit was larger than 5, if the unit digit was larger than 5, or if both digits were larger than 5. The participant was to respond NO if neither the decade digit nor the unit digit were larger than 5 (i.e., when both were smaller than 5). Again, presentation and timing were those used in Experiment 1. The responses were manual. The participant pressed the key "m" for a YES response and pressed the key "z" for a NO response.

Data analysis. In all analyses, RTs shorter than 150 ms or longer than 2,900 ms were excluded (a total of 1.8%, 1.9%, 4.4%, and 1.1% of the total number of trials for Participants 1–4, respectively). All RT analyses were based on correct responses, so error trials were also removed (9.7%, 8.3%, 3.0%, and 2.9% of the total number of trials, for Participants 1–4, respectively).

Results

The primary data of consequence (for model diagnosis) is the subset of double-target trials (factorial design B in Figure 2: Left Digit of Low or High Salience × Right Digit of Low or High Salience; see Table C1 in Appendix C). Each observer's RT data was analyzed separately by an ANOVA as shown in Table 4. The ANOVA results are consistent across all observers: The main effects of decade and unit were highly significant as was their interaction. This already provides evidence for a mean RT overadditive type of interaction. The interaction is shown graphically in Figure 6. Again, the forms of the functions are highly consistent across the observers. In terms of the underlying model, the initial evidence supports either minimum-time parallel race or interactive coactivation. Of the two alternatives, the Garner outcome in Experiment 1 already rules out independent parallel race. Nevertheless, we will seek further converging evidence for coactivation based on SFT analysis alone. Note also that the same initial results rule out serial processing or exhaustive processing (whether serial or parallel). In this instance, too, the exclusion of serial processing is already mandated by the Garner result of integrality (but see Townsend & Nozawa, 1997, on the possibility of dimensional interaction under exhaustive serial processing; see also Fitousi & Algom, 2018).

Table 4
Results of Analysis of Variance for Each of the Four Observers in Experiment 2

Observer	Source	*MSE*	*F*	*p*
1	Unit	13,048,989	227.8	.001***
	Decade	12,092,517	211.1	.001***
	Unit × Decade	6,493,425	113.4	.001***
2	Unit	4,078,054	87.59	.001***
	Decade	1,942,941	41.73	.001***
	Unit × Decade	551,946	11.86	.001***
3	Unit	14,527,565	90.53	.001***
	Decade	13,409,773	83.56	.001***
	Unit × Decade	4,801,360	29.92	.001***
4	Unit	12,748,194	167.86	.001***
	Decade	9,408,366	123.88	.001***
	Unit × Decade	5,148,297	67.79	.001***

*** $p < .001$.

To enable further unique specification of the underlying model, one must move from the means to the distributions. Therefore, we calculated the SIC(t) functions for each of the individual observers and these appear in Figure 7. Visual inspection shows that all functions are overwhelmingly positive, but also that each includes a small negative excursion at small values of t. The massive positivity provides convincing negative evidence against serial processing (of both self-termination and exhaustive) and parallel (exhaustive) processing—possibilities that are also inconsistent with the Garner outcome. Considering the magnitude of the negativity-positivity asymmetry in the SIC(t) functions as well as the location of the negativity blip, the functions are most consistent with a coactive type of processing.

The evidence provided by the confidence intervals in Figure 7 granted, we further applied a dedicated test developed by Houpt and Townsend (2010). Their procedure entails two separate assessments of the null hypothesis: One examines whether the largest positive value of the SIC is significantly different from zero and the second examines whether the largest negative value is significantly different from zero (in absolute terms). The test statistics is $\hat{D}+$, which is the supremum of these differences (see also, Houpt & Burns, 2017, p. 57). In the analyses, all positive parts of the SIC ($\hat{D}+$) for each of our four observers turned out to be significant (values of 0.22, 0.14, 0.20, and 0.19 for observers 1–4). Most important, all negative parts of the SIC ($\hat{D}-$) were also significant (values of 0.25, 0.18, 0.15, and 0.34, for observers 1–4). Therefore, the results of this test also support the presence of the small "blip" in the negative part of the SIC, a diagnostic feature of coactivation.

An important assumption underlying the use of the salience manipulation is that of *selective influence*. It means that manipulating the salience of the decade digits exclusively affects the (speed of) processing the decades and, conversely, that manipulating the salience of the unit digits exclusively affects the (speed of) processing the units. A necessary (though not sufficient) stipulation supporting the presence of selective influence is the RT ordering of the four possible decade-unit salience combinations

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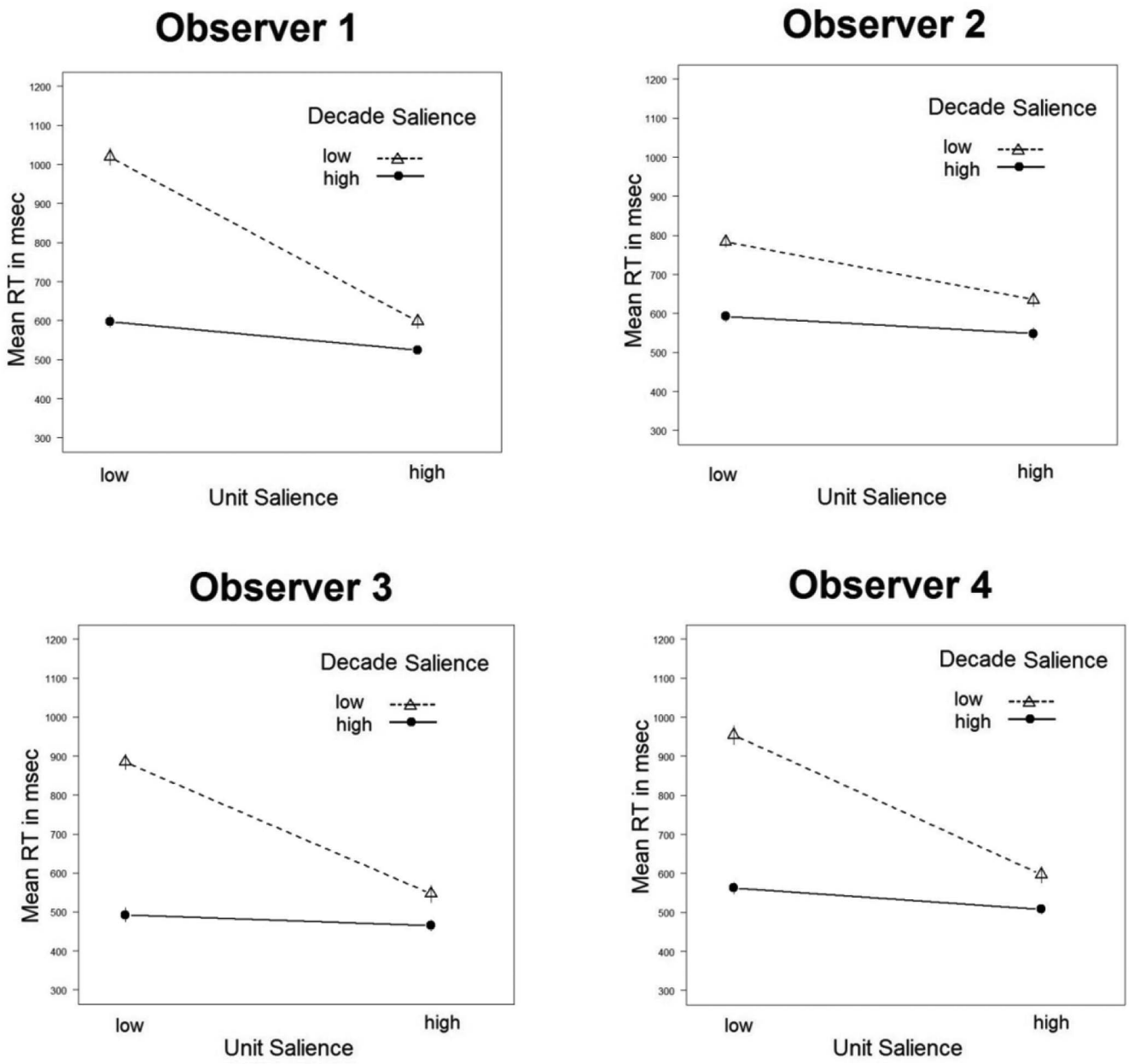


Figure 6. Factorial plots depicting the mean interaction contrast for four observers in Experiment 2. Error bars depict one standard error around the mean.

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such that HH $<$ HL, LH, $<$ LL. We present the respective survivor functions for each combination in Figure 8 and the results of the Kolmogorov–Smirnov tests performed on these survivor functions in Table 5. The predicted ordering is clearly satisfied by the functions in Figure 8 for each observer, consistent with the presence of selective influence.

Our final SFT analysis refers to changes in performance when the workload is increased from one to two targets. We considered three measures of performance in that context: Miller's inequality, Grice's inequality, and SFT's own capacity coefficient, $C(t)$. One can rearrange terms in the first two inequalities to express them on a common scale with $C(t)$ while retaining their original meaning (Townsend & Eidels, 2011).

Thus, the novel form of Miller's inequality can be written as,

$$C(t) \leq \frac{\log[S_D(t) + S_U(t) - 1]}{\log[S_D(t)S_U(t)]}$$

and Grice's inequality, too, can be transformed into:

$$C(t) \geq \frac{\log\{MIN[S_D(t), S_U(t)]\}}{\log[S_D(t)S_U(t)]}$$

again without a change in meaning.

In Figure 9, the three measures of capacity are presented on a common space, separately for each of the individual observers. In these plots, violations of the Miller bound are indicated when the capacity coefficient line is located higher than the Miller's bound line. Inspection of the functions reveals regions of t with SFT's C(t) greater than 1 with three of the four observers. These instances of supercapacity support coactivation. In other regions of *t*, C(t) was consistently below 1, but these capacity limitations were not severe enough to lead to violations of the (lower) Grice bound with any observer. As can be seen, the Miller bound was violated for Observers 1, 2 and 3, and, arguably for Observer 4. We also calculated the RTEs (with respect to the faster single-target) and these amounted to 256, 102, 207, and 143 ms, respectively, for Observers 1–4 (all RTEs were significant at $t \geq 8.50$, $p < .005$; see Appendix C for the full data). The outcome with the RTEs provides further evidence for coactivaton. Another test developed by Houpt and Townsend (2012; see also, Houpt & Burns, 2017) compares the observed RTs with those predicted by an unlimited capacity independent parallel model. We used this test, too, to evaluate the capacity coefficient functions. The z cores obtained were negative: −20.6, −19.3, −20.13, and −20.1 for Observers 1 to 4, respectively (with $p < .005$ in all cases). To recap, the processing model that most readily accommodates these data is coactivation.[3]

To summarize the results of the SFT analyses, they provide support for a coactive summation of information from the components of two-digit numbers. Serial models and all varieties of exhaustive processing are falsified. Now, coactivation is a quite extreme form of interaction and its presence in the current task of nonselective target detection is by no means trivial. After all, the task does not invite a global representation of the two-digit number; on the contrary, each digit is treated separately on equal footing as one entailing or not entailing a target. As far as the task goes, the stimulus is simply a pair of digits placed side-by-side without any implication of a shared property. The outcome by which the two digits coactive the response seems counterintuitive—until one realizes the overarching principle of the place-value system. This system is so deeply ingrained in people's psyche that it is in play even when the task does not invite any integration.

The finding of decade–unit interaction is also notable from the point of vantage of arithmetic. To perform the simplest arithmetic calculations, the component digits must preserve their separate autonomous values. Any interaction would compromise the correct arithmetic calculation and outcome. Nevertheless, in the face of digit independence in processing, a global representation also emerges. An example from music can serve to illustrate this point. Each individual musical note is carefully preserved and executed by the musician playing her instrument in the symphonic orchestra, yet a global representation, the melody, emerges. Arithmetic calculations share with the current tasks of Garner and SFT the request to attend and process the individual digits, yet the integrality and coactivation hint at the presence of a global representation. In the following discussion we propose a plausible processing model that incorporates both local and global action and representations within a well-defined time course.

General Discussion

We cannot overemphasize the omnipresence of the place-value principle whenever people are exposed to multidigit numbers for any purpose. This principle governs the simplest arithmetic operations, the structure of monetary sums (the same number 7 is worth

[3] We are aware of the related development of resilience functions (Little, Eidels, Fific, & Wang, 2015). The goal of resilience analysis is to test possible influence of distractors on processing of single targets. We do not apply this analysis in the present work on the grounds that is unsuited to the current stimuli, but also that the resilience development is vulnerable on multiple levels in general.

First consider the current situation. The ultimate test diagnosing an(y) influence of a distractor is experimental. Perform the same experiment once with single targets plus distractors and once with single targets alone (i.e., present the same target singly without the distractor). Then, comparing performance with and without the presence of a distractor (e.g., Fitousi & Algom, 2018) resolves the issue. The absence of a difference would show the absence of influence by the distractors. Recall also that, as a rule, experimental tests are always preferable to mathematical or statistical analysis. Notably however, an experimental control test is excluded in the present case. In the present case, single targets cannot be presented alone. One cannot just present a decade value on its own.

Second, consider the general case. The method suffers from several weaknesses. Below, we note just a couple.

1. Resilience analysis leads to the absurd result that diagnosis with respect to the capacity and architecture of the system can be made without including the double-target or redundant-targets condition in the experiment! This is why.
   The resilience difference function is the difference between two resilience functions [$R(t)$] with distractors of high (Y_H X_H) and low (Y_L X_L) discriminability:

$$R(t)_{diff} = R(t)_{high} - R(t)_{low} = \frac{H_{AB}}{H_{AY_H} + H_{X_HB}} - \frac{H_{AB}}{H_{AY_L} + H_{X_LB}},$$

   with H the integrated hazard function with respect to h and with A and B denoting the locations of the two targets. Note also that targets (locations), A and B, are associated with distractors Y and X, respectively.
   Because inferences with respect to the resilience difference function are based on their relations to a fixed quantity (e.g., >0), one can algebraically discard the redundant condition (H_{AB}) from computation. For example,

$$R(t)_{diff} = \frac{H_{AB}}{H_{AY_H} + H_{X_HB}} - \frac{H_{AB}}{H_{AY_L} + H_{X_LB}} > 0$$

$$\frac{H_{AB}}{H_{AY_H} + H_{X_HB}} > \frac{H_{AB}}{H_{AY_L} + H_{X_LB}}$$

$$\frac{1}{H_{AY_H} + H_{X_HB}} > \frac{1}{H_{AY_L} + H_{X_LB}}$$

   Again, the resilience difference function yields the inauspicious outcome that diagnosis with respect to architecture can be made without including the redundant target condition in the design. Even when the mathematical derivation is sound, the outcome is difficult to justify empirically. In the present case, this would mean that in a study focusing on two-digit numbers one does not present two-digit numbers to the subjects. It is difficult to advocate this analysis in experimental psychology.

2. The nature of the relations between the single signal and the distractor are not specified in an exhaustive fashion in the resilience development. First, the distractor need not impair signal processing; it can also amplify and support signal processing. In general, the signal-distractor relation should be subjected to the same capacity analysis, which is done with the system as a whole.

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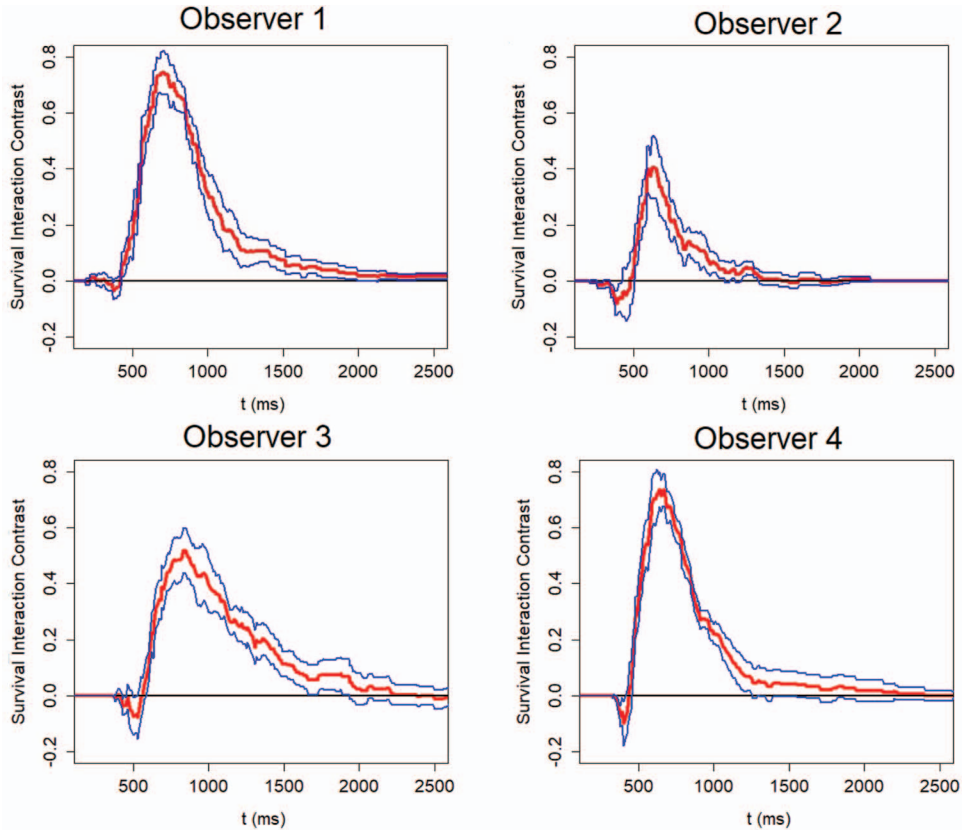


Figure 7. SIC(t) functions for four participants of Experiment 2 (blue line), with 95% confidence intervals (bootstrapped). The functions are most consistent with a coactive architecture. SIC = survivor interaction contrast. See the online article for the color version of this figure.

$70 when the price is $73 but is worth 7 cents when the price is $37), or, indeed, the pagination of this article. The essential feature of the place-value system is its inherent asymmetry. In multidigit numbers, the digit at the left(most) position counts more than any digit to its right. Considering our multiplicative system to the base 10, each place value of a given number is 10 times greater than that of the number to its right. The asymmetry is reflected in mathematical practice; for example, one typically rounds 73 to the nearest tens (70) rather to any particular unit. Although the leftmost numbers comprise the same 10 "Hindu-Arabic" numerals (0–9), they are nonetheless accorded special names: tens, hundreds, thousands, or millions. Considering logarithms, the most important part is their "characteristic" or the integer, whereas the fractional part, the "mantissa," counts less (the linguistic origin of "mantissa" is makeweight or worthless addition). Now, the principal outcome emanating from this study is the psychological reality of this mathematical device. The place-value structure, which is taught in elementary school as a user's guide, evolves over years of intensive use and mutates into a paramount rule of numerical cognition. The tools deployed in this study all converge on a single cognitive rule: Position-dependent processing of the components of two-digit numbers, which, in turn, determines their perception (see Anderson, 1982, for a discussion of the related concept of "cognitive algebra").

The asymmetric pattern found in the Garner experiment taps the psychological counterpart of the mathematical apparatus of place value. However, psychology goes beyond mathematics in this case. Our results show that people cannot attend exclusively to the unit digit of a two-digit number. They ineluctably notice the decade digit and this failure of full selectivity impacts performance in any task that involves units (e.g., the SFT task of target detection). In the language of Garner (1970, 1974a, 1974b, 1976; see also, Algom, 2016; Algom & Fitousi, 2016; Algom et al., 2017; Melara & Algom, 2003), the perception of units in a two-digit number is subject to integral processing whereby information from both digits combine to produce the response. In contrast, people are capable of focusing fully selectively on the decade digit of the two-digit number. In the language of Garner again, the perception of the left digit is separable so that processing is not influenced by the unit digits (unless the latter predict the former's value). A notable feature of these results is that the pairs of numerals in this study were *not* presented to the observer as two-digit numbers. The Garner paradigm in particular mandates responding only to a single, task-relevant component, the left or the right digit. Nevertheless, the decade-unit asymmetry still dominated performance. The upshot is, a "two-digit cloud" is always present when single numerals are presented side-by-side (even for separate assignments), and this cloud determines performance with each of the nominally single digits.

The last point is notable. We hypothesized the presence in the cognitive system of a two-digit structure despite the separate tasks and responses referring to each single digit alone. Do we have

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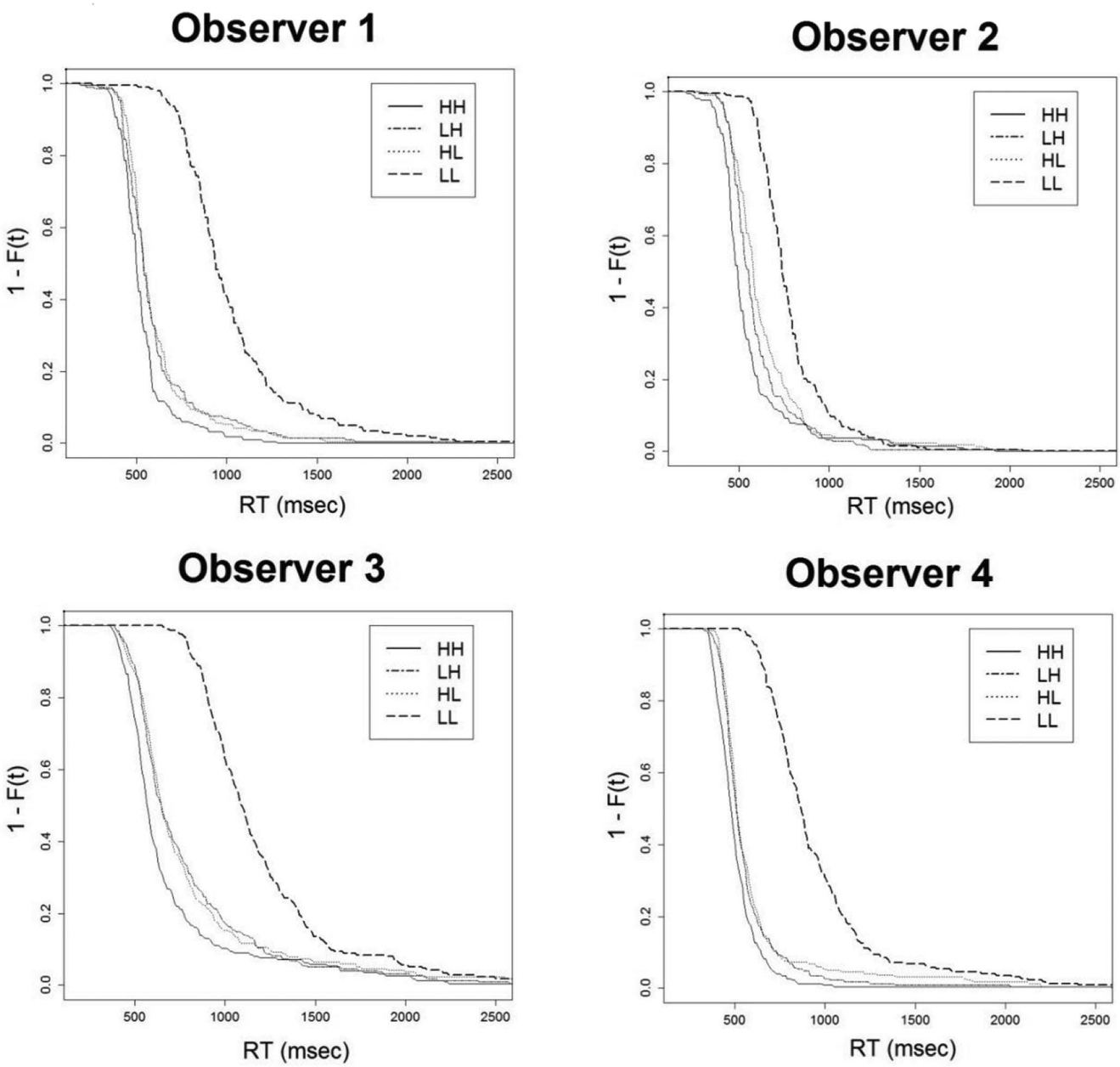


Figure 8. Survivor functions, $S(t)$, from the double-targets subset under the factorial salience manipulation. HH = decade of high salience, unit of high salience; LH = decade of low salience, unit of high salience; HL = decade of high salience, unit of low salience; LL = decade of low salience, unit of low salience.

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evidence for this hypothesis? To get a clue, we performed a novel analysis on the Garner data, treating each pair of digits as though it were a truly two-digit number (ignoring for the moment the presentation and tasks of the Garner design). We then examined the appearance of a distance effect with these fictional stimuli with respect to the boundary of 55 (the unpresented reference values for both components, hence a "two-digit number"). The results from the filtering condition (that included all numbers) are presented separately in Figure 10 for the left digit (presumably standing for decades) and the right digit (standing for units). Clearly, a global distance effect appeared for the left digit, r(RT, log distance) = −.74, $t(62) = -7.18$, $p < .005$, but not for the right digit, $r = .05$, $t(62) = 0.41$, $p = .67$. The former effect is similar to that observed by Dehaene et al. (1990) and Hinrichs et al. (1981) in tasks of comparative judgments with two-digit numbers. In those studies, the pairs of numerals were presented explicitly as ordinary two-digit numbers (unlike the present case). The outcome of this exercise supports the conjecture that, at some higher cognitive level, the participants represented the separate digits as an integrated two-digit number (the instructions to attend to each digit

Table 5
One Sided Kolmogorov–Smirnov D̂ Values Regarding the Survival Functions

Obs.	$S_{LL} > S_{HH}$	$S_{LL} > S_{HL}$	$S_{LL} > S_{LH}$	$S_{LH} > S_{HH}$	$S_{HL} > S_{HH}$
1	.86***	.79***	.77***	.20***	.22***
2	.71***	.50***	.60***	.25***	.34***
3	.78***	.63***	.66***	.21***	.24***
4	.82***	.75***	.76***	.16**	.20***

Note. The missing comparison is not necessary as the functions do not cross. Significant values mean that H_0 was refuted and that the prescribed ordering held. Note that the relations between S_{LH} and S_{HL} were not tested because their relation has no bearing on ordering.
** $p < .01$. *** $p < .001$.

singly notwithstanding). The nonuniform outcome underscores the deep asymmetry in processing between the digits standing for the decades and those standing for the units.

Next consider the SFT task: The observer reported the presence of a larger-than-5 digit in any place in the stimulus display. Thus, the two digits were treated on an equal footing as separate entities. This nonselective task arguably breaks up any provisional "two-digit cloud" even more than does the Garner task. It is on this background that the finding of an interactive model was surprising. Why should two putatively independent channels of information coactivate the response? After all, the stimulus *82* is not different from the stimulus *XO* as far as target detection is required (say the target of *8* in the first instance and the target of *O* in the second instance). Two targets—*88* and *OO*—yield faster YES responses than a single target, but in the case of the letters this RTE presumably derives from the statistical advantage accrued by the presence of multiple (independent) targets. In the case of the digits, by contrast, our SFT analyses show that performance was rooted in genuine interaction between the two numbers. Why should the two interact when any reference to the concept of "two-digit number" was neither made nor required? A glimpse at the literature helps to understand this point.

Coactivation in the literature is most often (exclusively?) found when there is a common goal or need shared by the separate pieces of information. Otherwise, there is no reason for separate channels to coalesce and coactivate the response. Suppose that a driver approaches an intersection on a dark and rainy night and must decide whether the traffic lights are red or green (one information channel) and whether pedestrians are crossing (a second information channel). Clearly, if the light is red or if pedestrians are

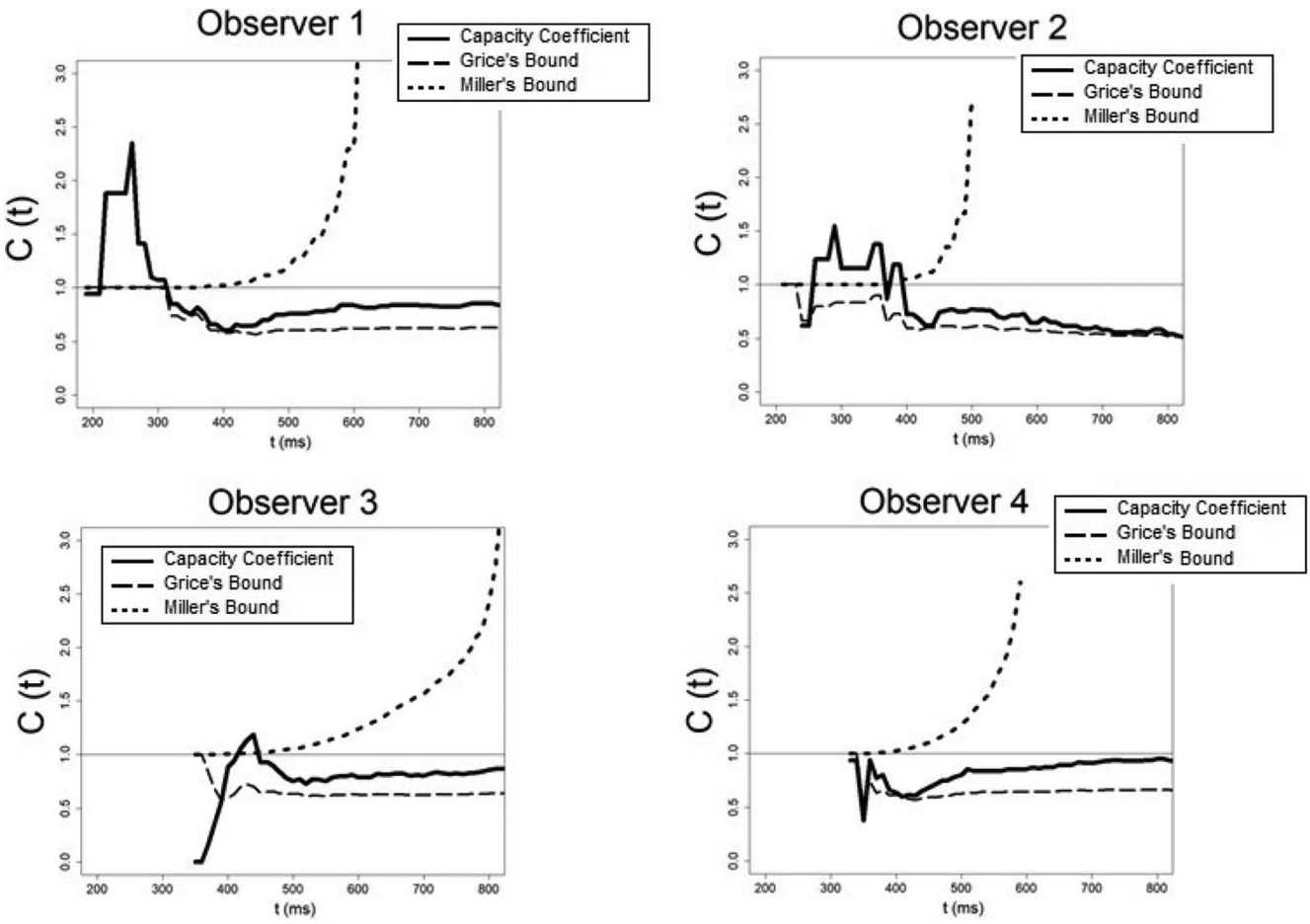


Figure 9. The Grice and the Miller bounds transformed onto the capacity coefficient space for four observers in Experiment 2.

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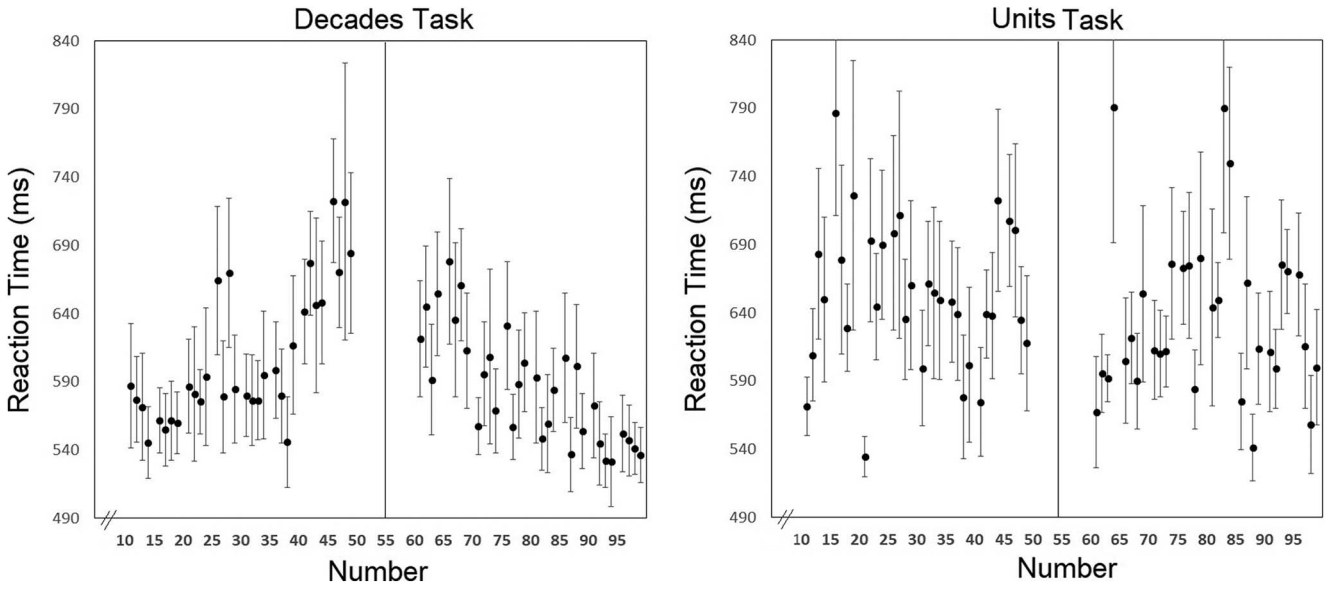


Figure 10. Reaction times (RTs) with the pairs of digits considered as genuine two-digit numbers. Left panel: RT to the left (decade) digit. Right panel: RT to the right (unit) digit. The bars depict one standard error around the mean.

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crossing, braking is needed. Clearly, too, when both targets are present—a red light and pedestrians crossing—the braking response will be faster than with each individual target alone (= RTE). Now true integration is expected in this case (beyond mere statistical facilitation) given that both stimuli are signals of danger (e.g., Bushmakin, Eidels, & Heathcote, 2017). While sailing in your boat in the lake, the double targets of thunder and lightning are integrated in your cognitive system to signal return to base. The two indicate the imminent storm more strongly than either target alone. In the laboratory, Miller (1982, 1986) used bimodal visual and auditory stimuli in divided attention designs and consistently found coactivation. The common goal or need in this case was obvious: to detect the mere presence of a stimulus. Is there a common conceptual umbrella in the current study of digit detection?

The clue is provided by the outcome of the Garner experiment and our attendant hypothesis. Although the SFT task lacks any obvious umbrella common to the detection of each digit, the assumption of a ubiquitous two-digit representation (whenever single digits are presented side-by-side) provides such a concept. Once the "two-digit number" concept is in force, the individual digits assume meaning, if implicitly, as decades and units. Now the Garner results showed that integral processing exists with the unit component, a component that is fully attended in every trial of the SFT divided attention task. So, an interactive model, coactivation, is the natural outcome of processing in the SFT task.

To recap, the two designs and attendant analyses complemented one another to converge on the same mode of processing. In particular, the combination in the SFT of architecture-oriented and capacity-oriented analyses helped us to identify the system as coactive (see Figure 3). In coactive processing, information from the two channels (left digit, right digit) is first processed in parallel, and then converges downstream to speed up the response. The current coactive model is actually a special case of an interactive-facilitatory model (Eidels et al., 2011). Considering the Garner outcome, the asymmetric pattern in fact mandated coactivation because integral dimensions (re the units) imply an interactive mode of processing. The two paradigms converged in falsifying parallel independent processing as well as general serial and exhaustive models.

Our discussion thus far focused on the psychological underpinnings of an ever-present global representation (even when the task at hand refers to each component digit alone). However, we would be remiss to ignore the excellent performance (in both the Garner and the SFT tasks) with respect to each single digit. As we also recounted, arithmetic operations also are based on single digit processing. And, the well-documented "decade-unit compatibility" effect (Nuerk et al., 2001) is symmetric (rather than asymmetric), which indicates that the single digits have not yet acquired meaning as components of a two-digit number. It is the crosstalk between single equal components that generates the (symmetric) compatibility effect. Note that the very designation, "*decade-unit* compatibility," might be misleading. These observations indicate that the standard coactive model might not tell the full story.

Therefore, we suggest a novel model, *the interactive race coactive model*, which is capable of accommodating all aspects of the available data. It is offered as a theoretic structure at this point. The proposed model is a conceptual compromise between the standard coactive model (see Figure 3) and the interactive race model suggested by Mordkoff and Yantis (1991, 1993). According to the Mordkoff and Yantis (1991) model, perceptual channels may exchange information. This assumption differs from the stipulation of independent race models that the channels be independent. However, in the Mordkoff and Yantis model the channels keep their separation throughout processing so that a separate decision is made by each channel (as in independent parallel race models).

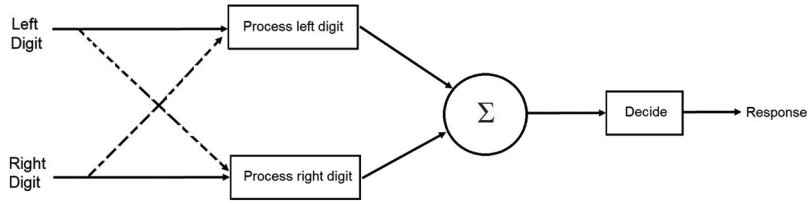


Figure 11. Schematics of the interactive race coactive model. There is a race between parallel processes, but they exchange information throughout activation. Eventually, the activations from the two channels are pooled prior to decision.

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This differs from the coactivation model, which assumes that the activations from both targets are pooled together prior to decision. Our own interactive race coactive model is distinct from both Mordkoff and Yantis's interactive race model and from the standard coactivation model. We suggest that at early stage of processing the two parallel channels (can) exchange information, but that later on the channels coalesce to produce a single decision and response (see Figure 11).

According to the interactive race coactive model, the left and the right digits are processed in parallel along separate channels up to a point in time when evidence from the two channels is merged together into a single representation. The compatibility effect is produced at an early stage when information is still separate, and the two digits have not yet accrued meaning as decade and unit. The conceptual structure of the two-digit number is imposed later and forces the merging of the channels and coactivation. The upshot is both single-digit and global representation are needed (see Verguts & De Moor, 2005, on this point). The former is needed in elementary arithmetic calculations (and in other tasks), the latter is needed to represent the overall quantity of the two-digit number.

When people meet two-digit numbers without time pressure, as in most everyday life, they perceive their magnitude based mainly, often exclusively, on the left digit standing for the decade. When overall magnitude is relevant (e.g., the price of a commodity), there is no need to decompose the number. Our Garner results tap the psychological underpinnings of this practice. In everyday parlance, say on the subject of income, people often omit (some of) the right-hand digits (as a harmless shorthand); for another example, just consider the title of the successful TV series, "Thirty-something." The units are perceived, but, obviously, they count less. We should also mention that the distance effect does not hold in nonspeeded responding (and, contrary to common belief, it is not robust even in speeded responding; e.g., Bar et al., 2019; Cohen & Quinlan, 2016).

Finally, a word is in order on the new interactive race coactive model. This model is distinct from the mainstream models discussed in the stochastic modeling literature (see, e.g., Figure 1 in Townsend & Nozawa, 1995, or Figure 1 in Eidels et al., 2011). We now use various simulations to elucidate its properties, but the model already accounts well for major phenomena of numerical cognition. It may also account for phenomena beyond number. Candidate domains entail the processing of letters including the phenomenon of "word superiority effect" (e.g., Arieh & Algom, 2002; Houpt, Townsend, & Donkin, 2014).

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Appendix A

Stimuli in Experiment 1

List of two-digit numbers used in Experiment 1, classified by task (unit, decade) and the type of Garner block (filtering, baseline, positive correlation, negative correlation)

Unit and decade tasks—filtering: 11, 12, 13, 14, 16, 17, 18, 19, 21, 22, 23, 24, 26, 27, 28, 29, 31, 32, 33, 34, 36, 37, 38, 39, 41, 42, 43, 44, 46, 47, 48, 49, 61, 62, 63, 64, 66, 67, 68, 69, 71, 72, 73, 74, 76, 77, 78, 79, 81, 82, 83, 84, 86, 87, 88, 89, 91, 92, 93, 94, 96, 97, 98, and 99.

Decade task—baseline (unit $<$ 5): 11, 12, 13, 14, 21, 22, 23, 24, 31, 32, 33, 34, 41, 42, 43, 44, 61, 62, 63, 64, 71, 72, 73, 74, 81, 82, 83, 84, 91, 92, 93, and 94.

Decade task—baseline (unit $>$ 5): 16, 17, 18, 19, 26, 27, 28, 29, 36, 37, 38, 39, 46, 47, 48, 49, 66, 67, 68, 69, 76, 77, 78, 79, 86, 87, 88, 89, 96, 97, 98, and 99.

Unit task—baseline (decade $<$ 5): 11, 12, 13, 14, 16, 17, 18, 19, 21, 22, 23, 24, 26, 27, 28, 29, 31, 32, 33, 34, 36, 37, 38, 39, 41, 42, 43, 44, 46, 47, 48, and 49.

Unit task—baseline (decade $>$ 5): 61, 62, 63, 64, 66, 67, 68, 69, 71, 72, 73, 74, 76, 77, 78, 79, 81, 82, 83, 84, 86, 87, 88, 89, 91, 92, 93, 94, 96, 97, 98, and 99.

Unit and decade task—positive correlation: 11, 12, 13, 14, 21, 22, 23, 24, 31, 32, 33, 34, 41, 42, 43, 44, 66, 67, 68, 69, 76, 77, 78, 79, 86, 87, 88, 89, 96, 97, 98, and 99.

Unit and decade task—negative correlation: 16, 17, 18, 19, 26, 27, 28, 29, 36, 37, 38, 39, 46, 47, 48, 49, 61, 62, 63, 64, 71, 72, 73, 74, 81, 82, 83, 84, 91, 92, 93, and 94.

Appendix B

Stimuli in Experiment 2

List of two-digit numbers used in Experiment 2 classified by experimental condition (double-target, single-target, non-target).

Single target trials: 16, 17, 18, 19, 26, 27, 28, 29, 36, 37, 38, 39, 46, 47, 48, 49, 61, 62, 63, 64, 71, 72, 73, 74, 81, 82, 83, 84, 91, 92, 93, and 94.

Double-target trials: 66, 67, 68, 69, 76, 77, 78, 79, 86, 87, 88, 89, 96, 97, 98, and 99.

No-target trials: 11, 12, 13, 14, 21, 22, 23, 24, 31, 32, 33, 34, 41, 42, 43, and 44.

(*Appendices continue*)

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Appendix C

Mean RTs and Error Rates in Experiment 2

Table C1
Summary of the Full Set of Data in Experiment 2: Mean Reaction Time (First Row) and Percent Error (Second Row) for Each Participant

					Single											
	Double				Unit				Decade				No target			
Obs.	HH	HL	LH	LL	HH	HL	LH	LL	HH	HL	LH	LL	HH	HL	LH	LL
1	525	597	598	1,019	544	1,215	591	1,065	566	602	1,203	1,059	618	1,278	1,182	1,132
	3.1	1.8	4.0	4.1	5.4	15.3	3.6	14.7	5.4	4.1	19.9	8.6	7.2	15.4	9.8	12.5
2	548	592	634	783	575	1,019	647	865	569	628	978	866	568	935	840	866
	2.7	2.7	.5	.9	5.4	21.2	4.5	11.3	11.4	1.3	33.6	13.6	7.5	6.5	4.9	6.8
3	688	782	792	1,194	788	1,470	822	1,355	704	761	1521	1,349	742	1433	1,435	1,403
	.4	.9	1.8	1.4	2.2	8.0	.9	3.3	5.8	1.8	8.6	1.9	2.3	3.4	1.3	3.3
4	508	562	596	954	513	1,224	582	1,049	523	574	1,189	999	556	1,172	1,055	999
	.4	.9	1.3	.4	3.1	5.5	2.2	7.7	2.7	.4	17.4	3.6	2.7	.5	1.4	2.4

Note. The letters H and L depict high salience and low salience, respectively, with the first letter denoting the salience of the decade digit and the second letter denoting the salience of the unit digit.

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